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Survey paper

Review of artificial intelligence applications in engineering design perspective

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ABSTRACT

Having passed the primitive phases and starting to revolutionize many different fields in some way, artificial intelligence is on its way to becoming a disruptive technology. It is also foreseen to totally change human-centred traditional engineering design approaches. Although still in the early phases, AI-powered engineering applications enable them to work with ambiguous design parameters and solve complex engineering problems, not otherwise possible with traditional design methods. This work attempts to shine a light on current progress and future research trends in AI applications in design/engineering design concepts, covering the last 15 years which is the ramp-up period for AI. Methods such as machine learning, genetic algorithm, and fuzzy logic have been carefully examined from an engineering design perspective. AI-powered design studies have been categorized and critically reviewed for various design stages such as inspiration, idea and concept generation, evaluation, optimization, decision-making, and modelling. As an overview result of this review, we can confidently say that the interest in data-based design methods and Explainable Artificial Intelligence (XAI) has increased in recent years. Furthermore, the use of AI methods in engineering design applications helps to obtain efficient, fast, accurate, and comprehensive results. Especially with deep learning methods and combinations, situations where human capacity is insufficient can be addressed efficiently. However, choosing the right AI method for a design problem under consideration is significantly important for such successful results. Hence, we have given an outline perspective on choosing the right AI method based on the literature outcomes for design problems.

1. Introduction

Traditional product and engineering design processes are human-centred and require expert knowledge involving scientific, intuitive, experiential, and creative methods (Pahl and Beitz, 1997). In recent years this conventional approach is changing with the use of Artificial intelligence (AI) in product and engineering design similar to all other engineering disciplines. AI is a general name for computer algorithms that imitate biotic mental processes or activities (Ertel, 2017; Liu et al., 2018). It can be used in tasks such as learning, understanding, estimating, problem-solving, suggestion, and decision-making in various disciplines including the engineering design process (Huang et al., 2019). The AI-supported design techniques can manage complex design operations such as comparison, evaluation, and estimation allowing the designer to focus on issues that require innovation and creativity. In this way, AI methods shorten the design processes, obtain precise results, and reduce overall design costs. Furthermore, it is superior in executing such tasks compared with humans due to its high computational ability, big data processing, and objective decision-making capabilities (Liao et al., 2020; Allison et al., 2022).

Of the many AI techniques, expert systems, fuzzy logic, artificial neural networks, and genetic algorithms are the most commonly used classical methods in the design evaluation and optimization processes for a very long time (Lu et al., 2012). However, the application of data-driven modern methods such as machine learning and deep learning in the design process has escalated in recent years. Various other AI techniques can be used in one or more stages of the design process such as idea inspiration, concept generation, evaluation, optimization, and decision-making (Allison et al., 2022). In a recent work by Saridakis and Dentsoras (2008), the use of classical AI techniques (i.e. fuzzy logic, genetic algorithm, and artificial neural network) in engineering design was reviewed. The research results have been a good implementation for the use of these methods in design problems. Similarly, Youssef et al. (2017) compared the implementation of such more traditional AI techniques in photovoltaic cell design and development. Chang

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et al. (2018) classified the learning methods applied in the defect detection and design of mechanical parts using machine learning and deep learning AI techniques. Abualigah and Diabat (2020) discussed the use of the Grasshopper optimization algorithm which is a meta-heuristic optimization algorithm in engineering design problems. They compiled the optimization algorithm results, the application types, and hybrid usage in design. Giri et al. (2019) conducted a systematic and comprehensive literature review on the use of AI techniques in the fashion and clothing industry. They determined the importance and scope of machine learning, deep learning, expert systems, decision support systems, and optimization algorithms applicable to this specific field. Specifically, the positive and negative aspects of using AI techniques in procurement, design, production, and distribution processes have been demonstrated. Verganti et al. (2020) examined the role of AI in innovation and design processes which is a mental process that requires experience and creativity. In another study, Nasiri and Khosravani (2021) investigated machine learning applications in the additive manufacturing field, optimizing the printing parameters, porosity, and defect detection, which have a great effect on mechanical properties. The place, importance, and the future of machine learning in 3D printing technologies were also discussed, and useful suggestions were given to researchers and practitioners. Lee (2021) examined the effect of using AI in product design from a social and economic sustainability perspective and presented the improvement of AI techniques in this regard by integrating the product design process into the product lifecycle.

Similar AI studies were published in architectural design and civil engineering fields as well as in engineering product design (Huang et al., 2019). Lu et al. (2012) conducted a comprehensive literature review on AI techniques applied in civil engineering. Current applications of AI methods such as evolutionary computation, fuzzy logic, expert systems, neural networks, and swarm intelligence were reviewed in their study. Salehi and Burgueño (2018) examined structural engineering applications of pattern recognition, machine learning, and deep learning methods. The advantages of these current AI methods over the aforementioned more traditional ones were discussed. Pan and Zhang (2021) conducted a detailed literature review on AI techniques used in civil engineering and construction management and their current applications, covering the period 1997–2020. They indicated that the combination of topics such as robotics, virtual reality, augmented reality, block chains, and 4D printers with AI will be potential research areas in the coming years.

Despite valuable contributions, previous review articles related to engineering and product design were performed with limited and narrow space. The classical AI techniques in some design problems and major advances in machine learning and deep learning applied to a product and engineering design in recent years were reviewed. However, almost all of these review studies and others in open literature deal solely with either classical or modern AI methods for specific applications. To be more precise, although the previous review articles have provided important information on the applications of artificial intelligence in engineering and product design, they did not consider traditional rule-based and modern data-dependent methods together. This work critically reviews the use of both classical (i.e. genetic algorithm, fuzzy logic, artificial neural networks) and modern AI techniques (i.e. machine learning and deep learning) in the scope of conceptual product design and hence complements the existing literature providing a more comprehensive framework combining the classical and modern AI techniques in the engineering design field. This way we aim to answer whether addressing these methods together can help discover more efficient and comprehensive AI solutions in design applications. Furthermore, the enormous increase in the use of artificial intelligence in engineering design, especially in the last 5 years (Statista, 2020), has necessitated this review. Another objective of this work is to classify artificial intelligence methods according to their use in engineering design processes (ideation, concept generation, evaluation, optimization, design support, modelling, etc.). Overall, our ultimate goal is to bring application areas of AI-assisted design as a whole to the attention of engineering designers, in the context of the framework outlined in the article. Considering the research questions, aims, and objectives given above, the current review has classified and evaluated the important works of the last 15 years. Both rule-based and data-driven AI methods applied to the conceptual design and embodiment design stages have been determined, examined, and compared with a view to future research projections. The strengths and weaknesses of the use of each AI method in design were examined.

The main contributions of this study to the literature can be summarized as follows: (1) Comprehensive AI applications in product and engineering design for the last 15 years have been examined, summarized, and classified. (2) The scope and limits of current practices have been determined. (3) Designers are guided in determining the appropriate AI method for the design problem. (4) The scope and directions of future AI studies in the design field are projected.

The remainder of this review paper was formed as follows; In Section 2, the systematic literature searches and paper review processes are explained. In Section 3, recent developments in AI methods and current applications are discussed. In Section 4, the AI methods within the scope of the study are briefly explained and sorted according to their current usage rates. In Section 5, AI applications in the engineering design processes are reviewed. In Section 6, the scope and limits of using AI in engineering design, and future research topics are discussed. Finally, the overall results of the study are presented in Section 7.

2. Systematic literature review

This study deals with the use of AI in engineering design processes. Since AI has a wide range of uses in design, the application of a systematic searching method is required to conduct a thorough, rational, and reproducible literature review. The review process here includes steps of searching, selecting, and obtaining information. Articles obtained from scientific databases are evaluated according to their content and scope, and those that are not within the scope are eliminated. The compiled papers are then classified according to their content.

Searching published papers

Published research papers were accessed through the two most used different scientific databases; "Scopus" and "Web of Science". The notations, formulas, and word strings used to search the databases and the number of papers found are shown in Table 1. All journal and conference papers found in the academic databases on AI and design between the years 2006–2021 were included in the literature research. English papers were reviewed without any country restrictions.

Selection of relevant articles

A four-stage elimination and selection process was followed to identify the publications to be excluded from around 2000 publications obtained in Table 1. In this context: (1) First of all, the same publications were removed from the joint papers pool. (2) In the next step, the paper title and keywords were examined, and publications that were outside the scope of the study were also eliminated. In the next stage, some papers were eliminated according to the abstract content. (4) After all these elimination processes, a total of 184 publications (132 of which were journals and 52 were conference papers) were determined and included in the study.

Extraction of information

In order to determine the subject and scope of the article without wasting time on unimportant details and exploratory literature, the review method was used (Blessing and Chakrabarti, 2009). The abstract of the papers was firstly reviewed, and the introduction and conclusion parts were only examined if the abstract was found to be relevant and interesting. If these sections were also related to the research topic, the findings/discussion sections were analysed. Publications included

Table 1
The notations, formulas, and word strings used in scientific databases and the resulting number of papers.

Database	Search words	Number of articles
Scopus:	TITLE-ABS-KEY ((“artificial intelligence” OR “decision support system” OR “data-driven design” OR “design automation” OR “machine learning” OR “design intelligence”) AND (“conceptual design” OR “engineering design” OR “mechanical design” OR “product design” OR “design development” OR “design optimization” OR “design assessment” OR “design evaluation” OR “design creativity” OR “idea generation”))	1221
Web of Science	(TS = ((“artificial intelligence” OR “decision support system” OR “data-driven design” OR “design automation” OR “machine learning” OR “design intelligence”) AND (“conceptual design” OR “engineering design” OR “mechanical design” OR “product design” OR “design development” OR “design optimization” OR “design assessment” OR “design evaluation” OR “design creativity” OR “idea generation”)) AND LANGUAGE: (English)	680

in the full review pool were the ones in line with the research topic in all aspects including the findings/discussion sections.

In the process of determining and classifying the purpose of the examined papers, answers were sought to the following questions;

- What kind of design problem did this study aim to solve?
- In which field was this study carried out?
- Why was this study needed?
- Which AI technique was used in problem-solving?
- What were the findings obtained?

3. An overview of AI methods

Artificial intelligence (AI) can be defined as computational methods that imitate human mental activities such as thinking, inference, classification, interpretation, estimation, and decision-making (Ertel, 2017; Neapolitan and Jiang, 2018; Salehi and Burgueño, 2018; Shehab et al., 2020). It is an interdisciplinary field of study combining information from several scientific fields such as mathematics, computers, neurology, biology, genetics, and engineering (Nti et al., 2021). AI research generally aims to perform human-specific mental tasks and activities quickly, effectively, accurately, and efficiently by computers (Lu et al., 2012; Rich, 1983). Its main difference from traditional software is that it can be processed with incomplete and uncertain data. That is, by creating semantic relationships between data, inferences for the past and predictions for the future can be made. It is now widely used not only in the design and engineering field, but also in the fields of education, health, railway traffic, economy, jurisdiction, and production (Shehab et al., 2020; Zhang et al., 2021).

Although AI technology, which has grown and become more complex in recent years, provides convenience in all areas of life, it is not known what reason the decisions taken are due to the black box working of the algorithms. Hence, the reliability of AI applications has come to the fore in various sectors where decisions are of great importance, especially in defence, health, and banking. The high-performing intelligent systems involving AI methods have to be transparent and explainable for them to be trusted in such applications. Only then confidence can be established in these actions if it is known how the decisions taken were reached (Minh et al., 2022; Vassiliades et al., 2021; Wells and Bednarz, 2021). Once the decision rationale is known, it can allow results to be verified, improved, and audited. However, the responsibility for the decisions made by artificial intelligence is unclear due to obscure internal processes. From this point of view, the concept of Explainable Artificial Intelligence (XAI) has emerged aiming to clarify the internal working structure of AI and the reasons for the decisions made. Even, human designers can learn from XAI-assisted designs in situations not yet been explored. XAI aims to understand cause–effect relationships, identify strengths and weaknesses, and figure out their future behaviour. Thus, transparent, explicable, and reliable human–machine cooperation can be achieved. Minh et al. (2022) divided XAI methods into three groups: pre-modelling explainability, interpretable model, and post-modelling explainability. Pre-modelling explainability is defined as gaining insight from the data after a series of data processing steps of the training dataset. Interpretable models are XAI methods that can be understood by examining the model structure and parameters by experts in the subject. Rule-based learning, linear/logistic regression, and decision trees are some of the XAI methods in these groups. Methods such as visual, textual, simplification, and attribute effect level of artificial intelligence decisions are accepted as post-modelling explainability. The main purpose of the methods in these groups is to identify the training performance of the inputs given to the model (Islam et al., 2022).

AI methods can be divided into two categories (i.e. strong and weak AI) according to problem-solving method employed. Strong AI techniques try to imitate the mental and heuristic processes that humans use when solving a problem (Lucci and Kopec, 2016; Verganti et al., 2020). The purpose of such systems is to give computers the ability to think, perceive, and reason. Weak AI, on the other hand, is generally computational software that successfully performs certain tasks but without involving heuristic processes. Although the word ‘weak’ may suggest otherwise, weak AI methods are usually more efficient and successful at a given specific task than a human. However, they cannot show the same success sustainably in any other tasks (Salehi and Burgueño, 2018; Verganti et al., 2020). It must be noted that most of the AI methods available in use today can be considered weak AI. Nevertheless, although still in the preliminary research phase, the concept of super AI has also emerged in recent years. This concept refers to software that has higher processing and thinking capabilities than humans.

Another purpose of this study is to define the most preferred AI methods and their areas of use in design processes. A keyword pool has been constituted, drawn from thousands of studies on the use of AI methods in design over the past 15 years (2006–2021). The list has been constructed by using word cloud format (Cidell, 2010) and frequency of use. It is understood that machine learning, genetic algorithm, artificial neural networks, fuzzy logic, swarm intelligence, data mining, and expert systems are the most repetition keywords among AI, respectively examining the list. Fig. 1 shows the number of publications on each of these AI methods between the years 2006–2021. The change in the interest in these AI methods over time can be clearly seen in the figure. First of all, the number of academic studies on the use of AI in design has increased rapidly since 2014. This can be partly attributed to the rapid development in computer technology and the ease of access to data stores. Machine learning and deep learning have a significant share in this increase. The interest in other AI techniques such as expert systems, fuzzy logic, artificial neural networks, and genetic algorithms is not as much, and the rate of usage over the years is actually decreasing in this case.

4. AI methods

Expert systems, fuzzy logic, artificial neural networks, and genetic algorithms are the most commonly used classical methods in the design evaluation and optimization processes for a very long time. However, the application of modern data-based methods such as machine learning and its sub-branch deep learning in the design process has increased in recent years. Various other AI techniques can be used in one or more stages of the design process such as idea inspiration, concept creation or generation, evaluation, optimization, and decision-making processes. Sometimes hybrid AI combining the methods such as

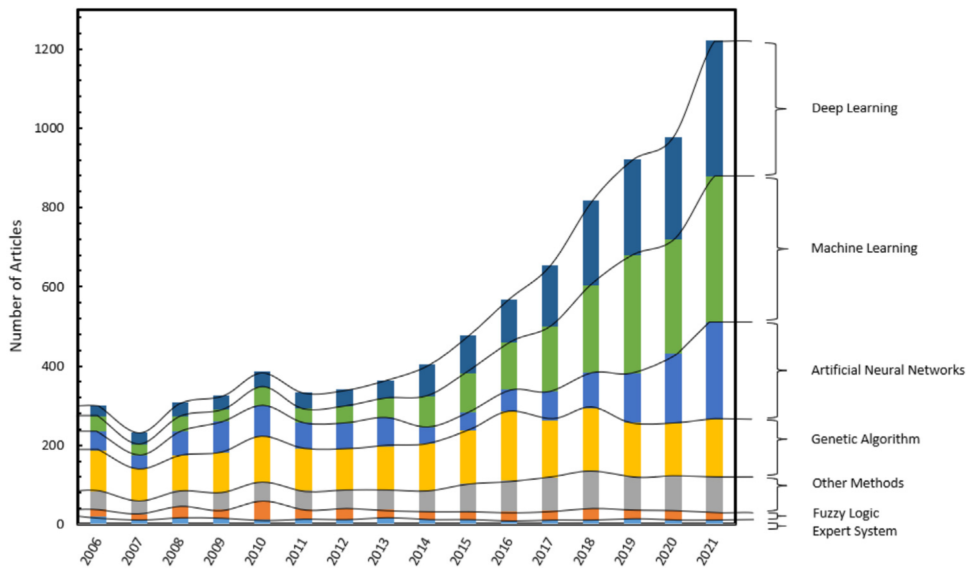

Fig. 1. Number of articles published per year on different AI methods specifically used in design applications between the years 2006 and 2021.

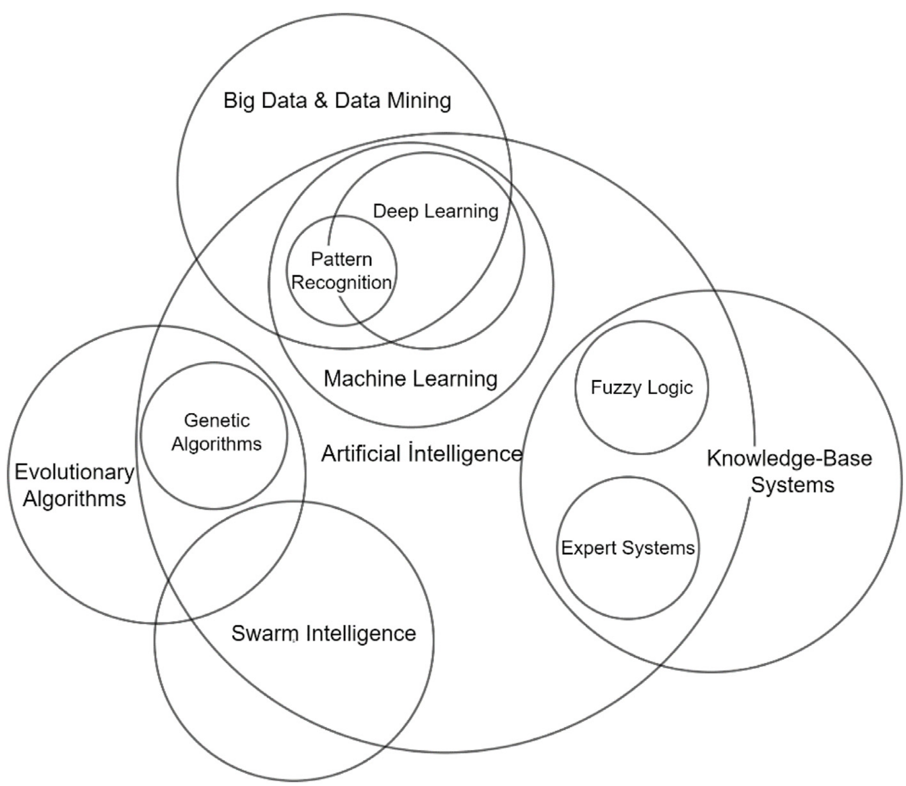

Fig. 2. Venn diagram showing an outline of relationships of the most widely used AI methods.

neural networks, fuzzy logic, and swarm intelligence can also be used simultaneously in some design applications (Zhang et al., 2021). The relationships of the many commonly used AI techniques are shown in Fig. 2. Since evolutionary algorithms, big data/mining, swarm intelligence, and knowledge-based systems can also be used in some other applications (such as classical computer programming and statistical computing), they were not classified as solely a sub-branch of AI in this figure. A brief introduction to these AI methods used in the design is presented in the remaining section.

4.1. Machine learning

Machine learning can be defined as statistical AI methods that can make predictions and inferences based on data (Alpaydın, 2014; Rebala et al., 2019; Salehi and Burgueño, 2018). This may sometimes be confused with traditional programming. However, machine learning does not need a clear algorithm to learn (Houssein et al., 2021) which is the case for programming. It rather detects similarities and patterns in any available data set and uses these to interpret future data. Hence,

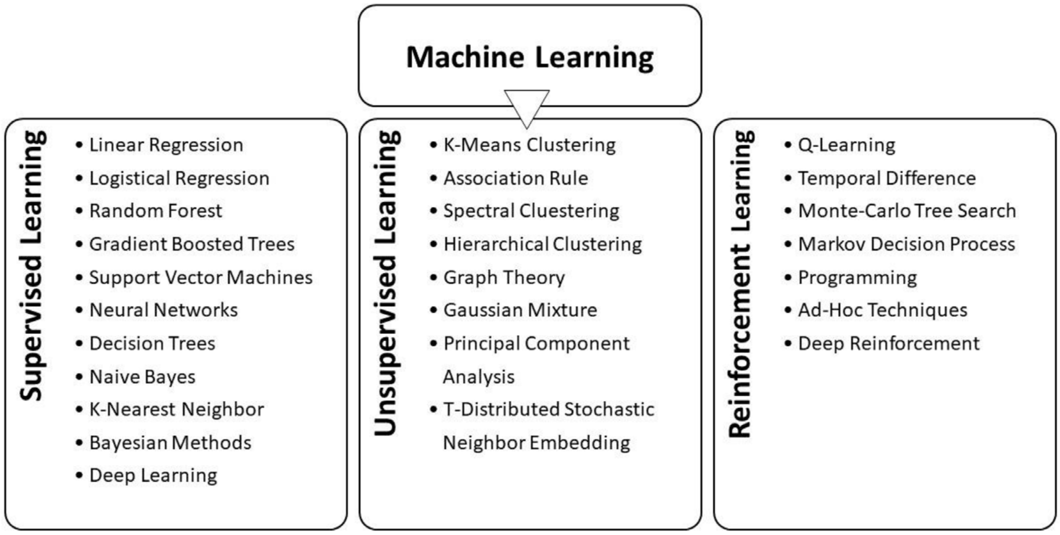

Fig. 3. The most commonly used machine learning algorithms categorized in accordance with the three different types of learning.

machine learning can be used in tasks such as natural language processing, sound and image, and pattern recognition, which are difficult to do with classical programming (Rebala et al., 2019). Statistical learning that is used to understand and interpret data is the basic theory behind modern machine learning algorithms (Harman and Kulkarni, 2011). Although statistical learning and machine learning are overlapping concepts, they have certain differences, especially in interpretability. The former is used in problems where there is a linear relationship between input data (independent variables) and output (dependent variables), such as linear regression and logistic regression. Since it can be determined how the changes in the independent variables will reflect on the dependent variable, it has a high explainable feature. In machine learning, on the other hand, the results are more difficult to interpret because of the lack of a linear relationship between the input and output variables. In some cases, interpretability can be ignored due to the ease of application of machine learning and its success in large data sets (Veribilimiokulu, 2018). These algorithms improve the ability to interpret with the increment of data, just like the human learning mechanism.

The increase in the amount of data (social media, online shopping sites, internet of things, photo and 3D model warehouses, etc.) and the ease of access to big data escalated the interest in the field of machine learning in the last 15 years. Moreover, super performance computers and the development of algorithms suitable for big data also contribute to and increase the success of learning, prediction, and identification. Thus, with machine learning algorithms, unique design concepts can be created that will increase user-product interaction and affect the user's purchasing preference (Chan et al., 2020; Han and Moghaddam, 2020; Singh and Tucker, 2017; Zhou et al., 2019). In addition, topology optimization can be performed with machine learning models instead of finite element analysis, which requires high computational power (Mukherjee et al., 2021). However, machine learning is often referred to as a black box because it is not fully explained how these algorithms arrive at their results. Broadly three different learning models are available in the concept of machine learning, namely; supervised, unsupervised, and reinforcement, as shown in Fig. 3.

4.1.1. Supervised learning

Supervised learning aims to obtain the best function (model) that describes the pattern between input and labelled output variables. In other words, learning takes place by determining the relationship between input values and label values (Alpaydın, 2014). With supervised learning, classification of outputs consisting of a number of categories (poor-average-good, female–male, acceptance–rejection, etc.) or regression tasks consisting of continuous numbers (air temperature, salary, weight, etc.) can be performed (Hügle et al., 2020). The datasets used are divided into two parts: train and testing (Jovel and Greiner, 2021). While the train data set is used for model feature selection and adjustment of network weights, the success of the machine learning model is measured with the test data set. It must be noted here that creating labelled datasets is cumbersome, and the input–output relationships between the data should be clear. This learning technique has a very high success rate with a minimum margin of error (Brynjolfsson and Mitchell, 2017). Artificial neural networks, logistic regression, linear regression, K-nearest neighbour, decision trees, and support vector machines are the most common supervised machine learning algorithms.

Artificial Neural Network (ANN)

The ANN mimics living nervous system structures (e.g. neurons, dendrites, and axons) and their relationships (Zhu et al., 2020). The nervous system and its features are reduced to a simple structure, and a mathematical operation model is created (Alanis et al., 2019; Dongare et al., 2012; Faul, 2020; Kumar et al., 2019). The most prominent feature of ANN is its ability to solve complex and unformulated problems via optimization of network weights between the interconnected neurons where the information is activated. This process, which is the basis of deep learning methods, is called network training. Although ANN is classified under the title of supervised learning in the literature, these networks and some deep learning algorithms can also be trained by unsupervised learning (Liu et al., 2022; Herzog and Suwelack, 2021).

There are many neural networks, some are more widely used than others. For example, a BackPropagation Neural Network (BPNN) is trained with feed-forward and back-propagation algorithms where information reaches the output layer and advances to the previous layers, respectively. In the former algorithm, the information from the previous layer is multiplied by the network weights in the current layer, and a bias is added to form a transfer function. Net input from this function is evaluated in the next stage activation function, and the information (net input) exceeding a given threshold value is transferred to the next layer. In the last step of this feed-forward algorithm, the difference between the value obtained in the output layer and the data label (e.g. 1-0) is called loss or error (Closs et al., 2008; Sang et al., 2021). The network weights are updated (trained) from the output layer to the input layer with the back propagation algorithm. The gradient descent algorithm used for this purpose reduces the difference between the target value and the actual value. The training time of neural networks depends on many parameters (number of neurons and layers, activation function, learning rate, initial weight values, etc.) that are difficult to predict (Rebala et al., 2019). In addition, high computer processing power may be required for network training. Another limitation of this method is the ambiguity in achieving results as it cannot be explained how they are obtained. In addition, for artificial neural networks to be used in the design problem, the design variables must be numerical or convertible to numerical data.

Artificial neural networks have many applications in the field of design as well as finance, banking, medicine, communication, marketing, and production sectors. This method is used in the product configuration creation, options evaluation, and optimization processes at the early design stage (Chen and Chiang, 2010; Feldhusen and Nagarajah, 2009; Lin and Yeh, 2015). Moreover, neural networks help designers to analyse customer needs and relate them to engineering features (Kutschenreiter-Praszkiewicz, 2013).

Deep learning

Deep learning is artificial neural networks that consist of multiple layers and can be trained using large data sets. The concept of depth refers to the number of layers of the neural network and the density of the learned information. Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Generative Adversarial Networks (GAN) are among the deep neural networks with the widest application area (Gal, 2016). Consisting of convolution, pooling, and fully connected layers, CNN is the most widely used feedforward deep neural network, especially in image processing. Here the repeating convolution and pooling layers (maybe hundreds of repeats) are used to filter and learn various information (attributes) from a given image (e.g. a product picture) and to reduce the image size by eliminating junk information with increased processing speed respectively. In the final stage of the CNN architecture, there is a neural network with fully connected flat layers and progressively reduced neurons converging an output solution. The matrix of pixels from the convolution and pooling layers are converted into a one-dimensional array by flattening for this final stage neural network data input. The number of learned features on visual data increases with the number of convolution layers. Here, unlike machine learning where the features to be learned is user-defined, CNN decides which feature to learn. This makes deep learning algorithms more successful in operations such as image classification, 2D and 3D object recognition, voice recognition, face recognition, and language translation (Houssein et al., 2021; Singh et al., 2019). Also, engineering design tasks such as topology optimization, generating unique design concepts, and even computer-aided engineering and simulation can be performed by deep learning methods (Oh et al., 2019). More details on such specific applications are given in the following sections. GAN, another deep learning method frequently used in design applications, has provided a new perspective on deep learning with its ability to perform tasks such as creating images that did not exist before, increasing the resolution of existing images, creating 3D models from 2D images, and estimating different angles of an object given an angle and text-to-image translation (Jin et al., 2020; Rahman, 2020). Two different networks (generative and discriminator) simultaneously available in a single architecture of the GAN enable to create unique data (Yi et al., 2019). However, this algorithm requires huge amounts of data for successful learning which may result in long processing times (Ray, 2019).

4.1.2. Unsupervised learning

Unsupervised learning algorithm enables the AI model to self-discover and detect data with common features from an unlabelled dataset and reveal hidden patterns among the data (Bertolini et al., 2021; Meng et al., 2020). Unlike supervised learning, there is no need for a supervisor in model training. However, it is difficult to objectively evaluate model performances (Ostberg et al., 2021). These algorithms are generally used to detect general data distribution, clustering, and size reduction. K-means clustering, association rules, auto-encoder, and principal component analysis (Abdi and Williams, 2010) are the most commonly used unsupervised learning algorithms. K-means clustering allows dividing data into meaningful groups by taking into account the distances between the data. On the other hand, association rules try to define relationships between data. Principal component analysis and auto-encoder enable size reduction to increase the processing speed and reduce unnecessary computational burden (Jovel and Greiner, 2021).

4.1.3. Reinforcement learning

Reinforced learning has a high future potential similar to the traditional trial and error method. Here a series of actions are taken to perform certain tasks. The actions that lead to the right result are rewarded (approved), and those that lead to the wrong result are punished (rejected) (Bertolini et al., 2021). For example, the energy and time consumption for a robot trying to reach its target in the shortest way can be considered as reward/punished grades. The main purpose of reinforcement learning is to develop high-level tactics by starting random trials. This method explores solutions on its own, but the developer needs to define clear goals and set rewards and penalties (Rebala et al., 2019). When all this is taken into account, the reinforcement learning model has the highest creativity potential as it can find different ways to solve a problem.

4.2. Genetic Algorithm

Genetic Algorithm (GA), developed by John Holland in the early 1970s, is a heuristic search algorithm that mimics the evolutionary mechanism of natural selection where strong individuals survive and weak die (Kramer, 2017; Kumar et al., 2010). Many terms (gene, chromosome, population, crossover, mutation, etc.) used in the GA system are adapted from natural systems. The candidate solutions (i.e. chromosomes in genetic terms) are composed of "gene" sequences showing solution properties. The genetic operators here, such as selection, crossover, and mutation are used to identify the best option in the solution space via a number of iterative steps (Gen et al., 2008; Tam et al., 2019).

The initial solution space (also called population) may consist of randomly selected tens or thousands of homogeneous chromosomes depending on the complexity of the problem (Roy and Chakraborty, 2013). The solution value of each chromosome is calculated with the fitness function which is then compared to the objective function (i.e. the desired solution value). Reproduction of new (offspring) chromosomes is ensured by the pairing of chromosomes with high fitness in the solution space (Kanagaraj et al., 2014; Lin et al., 2015). Only promising solutions are allowed to survive removing weak chromosomes from the population. This is also necessary to keep the population number constant and improve the solution quality (Khuat and Le, 2017). In order to increase genetic diversity, a limited number of crossover or mutation operations are applied between chromosome pairs. The latter refers to the modification of one or more genes on the chromosome within certain rules. Crossover and mutation processes are repeated continuously to achieve good results. On reaching the targeted number of generations, objective value, or a certain degree of improvement from the initial state, the GA operation is terminated. GA can achieve successful results, especially in large solution spaces and complex problems. However, reaching the global best depends on parameters such as population size, mutation rate, crossover method, and selection method. Also, the coding of chromosomes requires intensive user experience. Although it is an expensive calculation method, the quality of the final solution may not be at the desired level.

The GA is widely used in optimization problems such as planning (e.g. robot path), navigation (e.g. travelling salesman problem), layout (e.g. factory layout), and search (e.g. strategy creation). In the conceptual design phase of engineering design, which is the main focus of this work, GA can be used in the creation of design options, the decision-making, and evaluation phases (Han et al., 2021; Zheng et al., 2016). Apart from these, GA can be used for design tasks such as product form and colour determination, assembly sequence planning, and component placement (Fung et al., 2012). Preferring GA instead of classical methods to evaluate thousands of conceptual solutions saves time and labour.

4.3. Fuzzy logic

Fuzzy logic is a theory created by extending classical/binary logic. It is used in the mathematical modelling of ambiguous expressions (short, long, very long, etc.) analogous to human thinking systems and real-life complex problems where classical/binary logic falls short (de Aguiar et al., 2018; Hooda and Raich, 2017; Trautmann et al., 2020). The fuzzy logic system was developed by Lotfi Zadeh in the mid-1960s (Bělohlávek et al., 2017).

Unlike the classical 'one' and 'zero' binary categorical sets, the belonging of an element to a categorical set is determined by the degree of membership in fuzzy sets. This means an element can be a member of two different sets to a certain degree at the same time in fuzzy logic (de Aguiar et al., 2018). The degree of membership is expressed with values between 0 and 1 (Sutono et al., 2016). The fuzzy logic system consists of four main units: rule base, fuzzifier, inference mechanism, and defuzzifier. The rule base is an "If-Then" formatted information repository that gives the system the ability to reason. Fuzzification involves transforming the numerical values into membership functions such as very few, few, fast, slow, etc. (Babaei and Sheidaii, 2017; Sutono et al., 2016). In the inference mechanism, fuzzy input values are interpreted through the rule base, and fuzzy results are obtained. With the defuzzification process, the fuzzy results are converted into meaningful numerical values (Hooda and Raich, 2017).

The success of the system depends on parameters such as the number of input and output variables, the number of membership functions, the inference method (e.g. Mamdani, Sugeno, etc.), and the defuzzification method (e.g. weight averages, the centre of the area, etc.). An increase in the number of input and output variables also requires an increase in the number of rules. Also, creating membership functions and building a rule base may require significant user experience.

Fuzzy logic systems have a wide range of use in industry, such as automation, control systems, robotics, medicine, economy, sales and marketing, management, environment, and engineering. In the context of the conceptual design processes such as the process of evaluating design variants in line with user expectations, making decisions, product life cycle evaluation, and new product development process management have also successfully been used (Huang et al., 2013; Lanzotti et al., 2018; Bajaras and Agard, 2015; Ko, 2010a; Meng et al., 2015; Sousa-Zomer and Miguel, 2017; Trautmann et al., 2020).

5. AI applications in design

In this section, AI applications used in various design stages have been classified according to their intended use. In this context, applications in design tasks such as inspiration, idea and concept generation, concept evaluation, optimization, decision making, and modelling are considered (Liao et al., 2020).

5.1. Inspiration

One of the most critical stages of product design is idea generation. However, it is not always possible to generate creative and innovative ideas due to psychological inertia (design fixation) (Han et al., 2018a). Human designers can easily get stuck on some design ideas blocking the process of thinking towards creating new ones during the conceptual design process (Jansson and Smith, 1991; Vasconcelos and Crilly, 2016). This is a major obstacle to achieving the ideal/perfect design. The inspiration process is a complex one that requires extensive market research as well as domain expertise, intuition, a bright mind, and creative skills. Prolonging this process may increase product costs and decrease market share. AI has been used to overcome this obstacle via verbal, written, and visual inspirations (Goldschmidt and Sever, 2011; Goldschmidt and Smolkov, 2006; Jiang et al., 2021). Patents are perfect examples of the latter two inspirations. Technical documents containing technology and design information such as the technical functionalities, materials, working principles, and concepts in the patent content can be used as an important source of inspiration for design and product development (Fu et al., 2013; Mukherjea et al., 2005; Sarica et al., 2020). Hao et al. (2019a,b) tried to obtain inspiring functional terms from patent databases with the non-dominated sorting genetic algorithm II (NSGA-II) to create novel and applicable ideas/concepts. The effectiveness and applicability of the method are demonstrated for an innovative fan design problem. Sarica et al. (2020) developed a solely technology and design-based semantic network (TechNet) using the U.S. Patent Database (USPTO), Fig. 4. Semantic networks are an effective tool for finding and visualizing related concepts. Here, information is obtained from the technical terms used in patent data with natural language processing techniques followed by combining related concepts with a machine learning algorithm. Links between the terms are then determined as the distance between the terms affects the formation of creative ideas. The greater the distance, the greater the possibility of generating creative ideas. However, this proportionality is just the opposite as for the viability of the ideas, and hence there is always a compromise between the two. Liu et al. (2020a) developed a function-based patent information retrieval tool. Functional terms were classified using a semi-supervised Naive-Bayes machine learning algorithm. Term contradictions between different fields were avoided/reduced when patent information is abstracted and transformed into functions. Luo et al. (2018) created a Technology Space Map (TSM) by classifying more than 5 million patents according to the technologies used in the patents. Here technologies in which similar approaches are applied are described as "close", and technologies that are independent of each other are described as "far". Information was then combined or transferred between technologies at different distances enabling idea generation in open-ended ambiguous problems involving uncertainty.

Some other researchers used scientific literature and web resources to inspire. Liu et al. (2020a,b) created a data-driven concept network by identifying critically important conceptual words from web sources alleviating the burden of the literature review at the early design stage. Similarly, to the aforementioned semantic network, this machine learning-based approach also allows distance measurement between concepts, and the creativity levels are measured by detecting close-far relationships (Han et al., 2020a,b,c). Using a similar approach, Han et al. (2018a) developed a software "The Combinator", to combine ideas that are not directly related. Data obtained through web browsers are analysed with natural language processing tools and associated with semantic networks. The authors demonstrated some creative product ideas with unique combinations of similar concepts. Here, the new idea generation is triggered by combining image and text data in different forms providing convenience to designers in exploring the large solution space (Boden, 2004; Ward and Kolomyts, 2010). In another work, Jin and Dong (2020) inspired by the 998 RedDot award-winning products, analysed their design principles using data analysis software (i.e. QSR NVivo) and created 10 new design heuristics cards (Yilmaz et al., 2016). Some other researchers used ontology-based methods to reveal relationships between concepts with machine learning, text mining, and natural language processing tools. Shi et al. (2017) developed an unsupervised ontology network that focuses on design and engineering called WordNet. Textual data were obtained from academic journals and design and engineering sites using data mining techniques. WordNet is built with a simplified natural language processing tool and semantic networks. It represents the relationship between concepts in design and engineering with an ontology network.

Another rich data source which a few researchers were inspired are social networks such as Facebook, Twitter, LinkedIn, etc. where billions

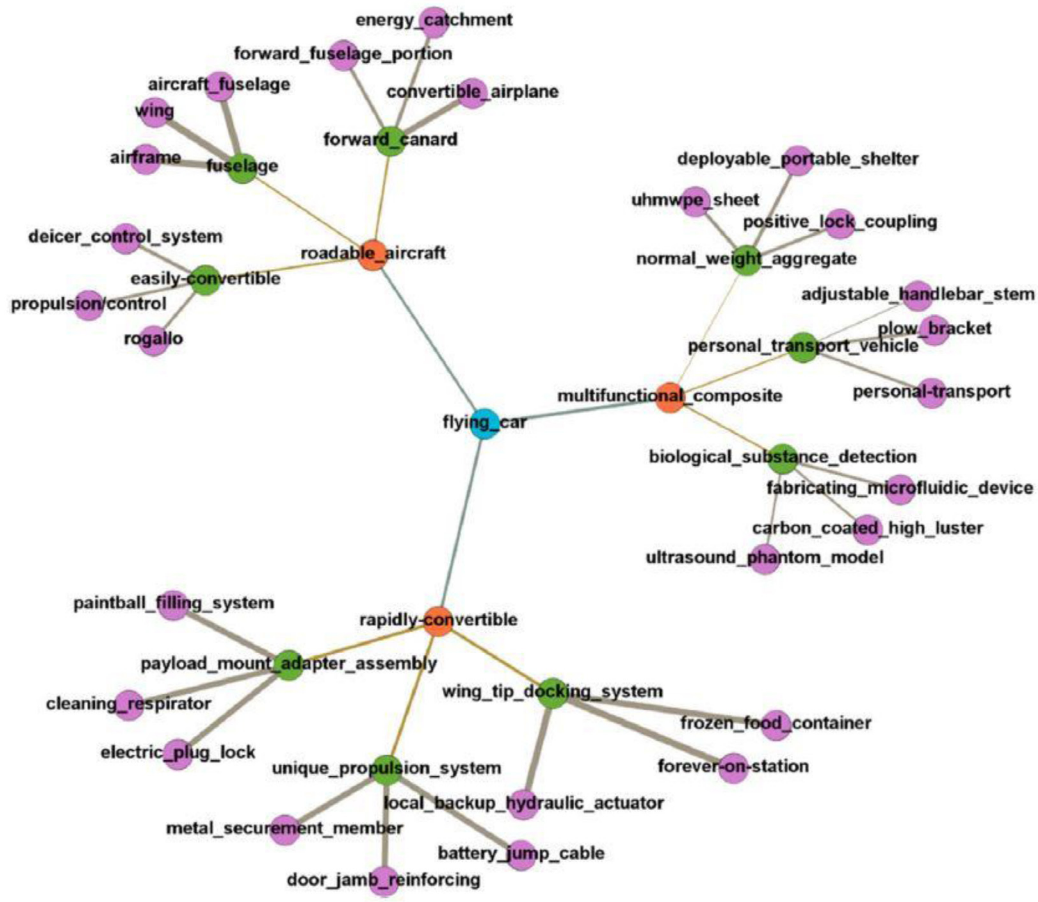

Fig. 4. "Flying car" concept network created with technology and design-based semantic network (TechNet) using the U.S. Patent Database (USPTO) (Sarica et al., 2020).

of people actively generate data. Han et al. (2020a,b,c) developed a computational approach to the acquisition, processing, and use of social media data to inspire and increase the creativity of designers in product design processes. Keywords from Twitter posts were collected with a commercial text-mining tool. Then numerical values were assigned to the obtained concepts according to their relationship with the keyword. This is followed by data analysis by natural language processing to inspire designers. In another study, Han and Moghaddam (2020) created a deep language model to identify customer needs from social network sites. This work is based on the BERT-NER model trained by Google AI Language. The BERT-NER is a pre-trained natural language process model on large text data and produces a field-independent result enabling to identify customer needs with very little manual data labelling.

Deep learning methods in inspiration applications have escalated in recent years. Bell and Bala used real product photos to train Siamese CNN (Bell and Bala, 2015). It creates a connotation for the target product by finding high-resolution images similar to the object in the photograph (armchair, cabinet, lampshade, etc.). Yu et al. (2019a,b) developed an automated design approach inspired by biological objects. Here, a CycleGAN architecture was developed to create the DesignGAN model. The DesignGANs model is trained with the Quick Draw dataset, which includes drawings of biological objects (hat, lamp, vase, etc.) and the target objects (octopus, flower, pineapple, etc.). Thus, the network can obtain unique drawings that contain the properties of biological objects. Chen et al. (2019) tried to create inspiration for designers with two different models. The first model is the aforementioned semantic network created with data mining and natural language processing techniques. The second one is an inspiration generation model based on the visual concept combination. Here, the GANs model, trained with the biological object and target object images, produces new images consisting of the combination of these two objects. An example is given in Fig. 5 where GAN is trained with spoon and leaf images to come up with a unique design concept inspired by nature. This way unique concept products containing both biological and target object properties can be produced. It can therefore be regarded as an effective way to create new inspirations/associations in the mind of the designer. Besides product design, this visual combination model can also be used for advertising and general idea generation.

To sum up, inspirational researches in the literature generally involve big data mining, machine learning, deep learning, and natural language processing algorithms. The widespread use of crowdsourcing tools facilitates access to the necessary data and information for this purpose. All inspiration methods mentioned here enable new and creative ideas to emerge, and new associations of these ideas can be obtained with different combinations of verbal, written, and visual data.

5.2. Idea and concept generation

The concept generation process begins with the identification of the problem in the early design phase (Han et al., 2018b). In this process, many criteria such as user demand, cost, aesthetics, ergonomics, functionality, production method and recycling should be taken into account. The main challenge of conceptual design is to produce concepts that balance these criteria. Conventionally generation of unique, creative, and useful concepts is a cumbersome process and requires a creative mind as well as knowledge and experience (Boden, 2004; Childs, 2014). Besides idea generation via inspiration covered in the

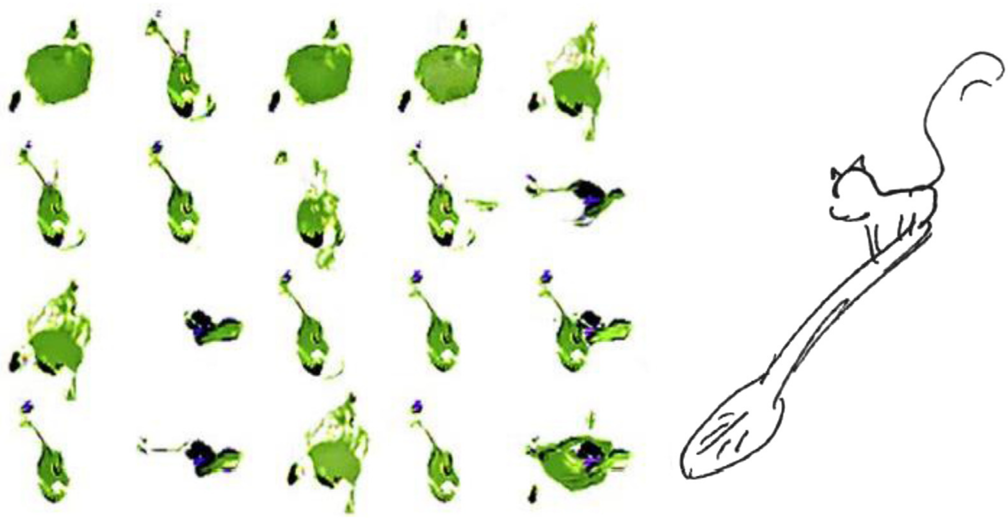

Fig. 5. Images produced by the GAN model trained with spoon and leaf images (left) to produce a unique design concept inspired by nature (right) (Chen et al., 2019).

previous section, AI can also be used for direct concept generation. Especially machine and deep learning methods help designers in creative concept generation (Yuan and Moghaddam, 2020; Zhao et al., 2021). Genetic algorithms can also be sometimes used in design solution space exploration and concept creation although it is classed as an optimization technique. For example, Matei used GA to explore the product configuration space and generate solution options in product designs (Matei and Contras, 2016; Matei, 2015). In another study by Lin et al. (2009), product configuration application was performed with an artificial neural network which is also an effective tool to determine ideal design configuration according to customer preferences. They used a morphological card (Börklü et al., 2018) consisting of product form elements to obtain all possible concept solutions. The numerical relationship between product appearance and product form elements here has been obtained with the Kansei Engineering approach (Shen and Wang, 2016). Chen and Chiang (2010) also developed a neural network model that takes into account customer expectations to generate not only fast and effective product concepts but also ideal product configurations. Lin and Chen (2016) conducted a user-cantered study in determining the ideal perfume bottle form using an artificial neural network to generate product form combinations. Camburn et al. (2019) worked on identifying new design concepts with artificial neural networks and crowdsourcing. Design concept ideas were identified in the datasets created with the spoken language. These design ideas have been classified according to their level of detail and clarity. This method can help to quickly scan large datasets (survey, competition, patent, etc.) and identify ideas. With the artificial neural network, product form features, and colour harmony that add aesthetic value to the product can also be determined. This is an important aspect of product design as some product form features create an aesthetic perception in the human mind. An exemplary work by Soni et al. (2013) demonstrated this using both ANN and GA to develop a basic design formulation for aesthetic products. Here, product form features determined by ANN using survey data are then used by the GA algorithm to generate alternative aesthetic design concepts. Fig. 6 shows the concept designs created this way for mixers, coffee makers, and washing machine products (Soni et al., 2013). Three design options have been generated for the end-user products in the figure concerning the product aesthetics. However, the aesthetic quality of the products is determined not only by the rough geometric features, but also by the material, texture, colour, and brand identity. Therefore, the created design options may need to be edited or even redesigned for real-life applications.

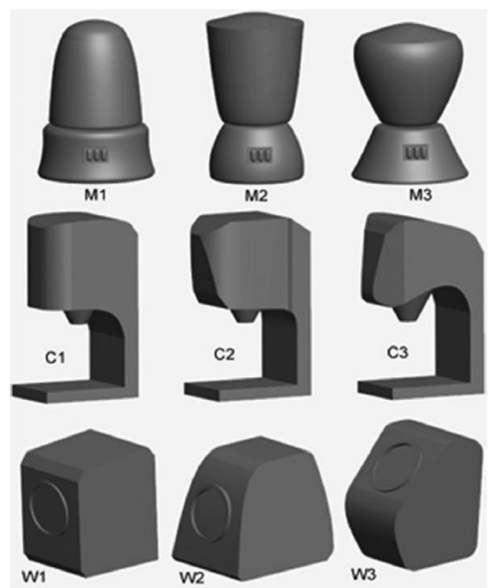

Fig. 6. Concept designs produced by the collaboration of an artificial neural network and genetic algorithm (Soni et al., 2013). Mixer (M), coffee maker (C), and washing machine (W).

Machine learning and data mining have also been successfully used for idea generation. These methods are especially effective to obtain hidden and explicit ideas from text and images. Large data sets such as patent databases, academic journals, web pages, and survey and speech data can be searched, and ideas from these sources are extracted and classified (Han and Moghaddam, 2020; Shi et al., 2017). For example, Zhou et al. (2019) conducted a customer needs analysis based on a large data source from online surveys, product reviews, and complaints using the machine learning method. This study is important because it uses a combination of supervised (i.e. fastText) and unsupervised (i.e. VADER) machine learning algorithms; the "fastText" algorithm was used to eliminate irrelevant data, and VADER (Valence Aware Dictionary and Sentiment Reasoner) was used to identify and classify customer needs. Bae and Kim (2011) generated new ideas in product design by identifying customer needs with a decision tree which is another machine learning algorithm. Using this approach, the authors aim to reduce production costs, increase product quality, shorten new product development processes, and increase competitiveness.

Deep learning algorithms can also be used to generate new product options by using the concept of computer vision to recognize, interpret,

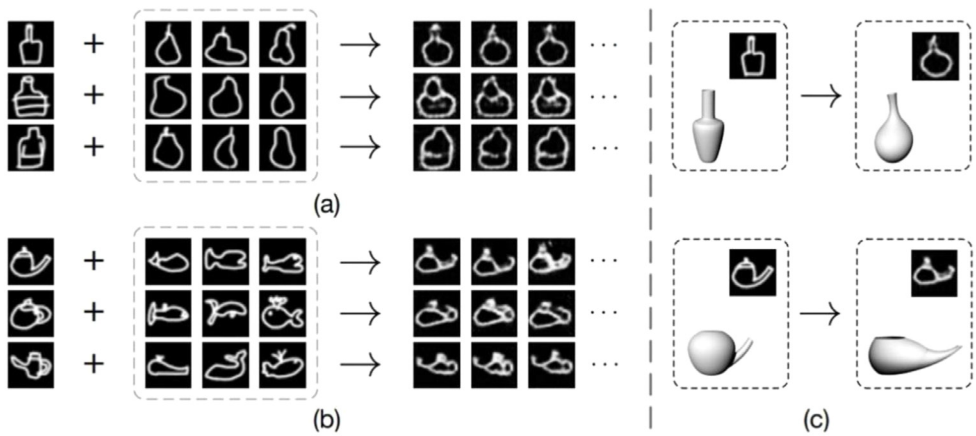

Fig. 7. Design concepts generated by the DesignGAN model (a) Biologically inspired bottle concepts produced with bottle and pear images (b) Biologically inspired design concepts produced with images of teapot and fish (c) Conversion of generated concepts to 3D models by human designers (Yu et al., 2019a).

and analyse visual data such as photos and videos. An unlimited number of new design options was demonstrated with deep learning generative models such as GANs and Variational Auto-Encoders (VAE) (Oh et al., 2019; Yonekura and Suzuki, 2021). Raina et al. (2019), imitated human design strategies in the idea and concept generation process with the deep learning method. The black box idea generation processes of human designers involving issues such as strategic and abstract thinking, and creativity were implemented in design problems where rules and strategies are ambiguous. This study proposes an image processing-based approach to achieve this goal. In this work, an auto-encoder was used to obtain design information from several consecutive images of the design stages implemented by human designers. An unsupervised neural network was then used to generate some unique design proposals.

In another study by Yu et al. (2019a,b) this deep learning algorithm (i.e. the GAN model) was used to develop a generative design model inspired by biological objects. Several biologically inspired images were created by combining the images of the target and biological objects (Fig. 7). The illustration shows two different case studies: a pear-inspired bottle and a fish-inspired teapot design. Here, a designer also created 3D models from the generated concepts. Oh et al. (2019) integrated topology optimization into the GAN algorithm to achieve aesthetic product designs with high engineering performance. A large amount of product image data obtained from the topology optimization process was used to train GANs models and create successful concept products. Topology optimization here enables multiplying the available data to increase the diversity and quality of the training set. Fig. 8 shows some exemplary concept models appealing not only from an aesthetic point of view but also from engineering performance, as claimed in this study. However, design viability from the engineering point of view was not tested on 3D concept products.

Another GAN algorithm (AttributeGAN or AttGAN) based work attempted to develop concepts keeping some attributes from the dataset unchanged and to edit only the other features (Mirza and Osindero, 2014). However, although this model achieved successful results in editing and producing face photos, it could not achieve the same success in other datasets. To cover this problem, Yuan and Moghaddam (2020) further developed the Attribute GAN (called Design Attribute GAN or DAGAN) model for textile products by using a special loss function and a discriminator loss in the AttGAN model. The aim here was again to organize only the specified features in the visual data set containing fashion products (e.g. vest, t-short) to produce unique concept fashion products with desired features unchanged. It should be noted here that although the latter modified AttGAN for a specific design problem was successful, it is still not a global solution to the aforementioned problem, and hence each different application requires such modifications. In another study by Chen and Ahmed (2020), performance-augmented diverse GAN (PaDGAN) models using a new probabilistic point process-based loss function are proposed to improve the performance of the GAN model. It is stated that this model expands the design area for quality solutions in design examples (e.g. aerofoil design) and increases the innovation and diversity of the designs produced. Another interesting work in this context proposed a neuroscientific method to obtain product concepts with user-desired features (Wang et al., 2020). Electroencephalography (EEG) signals were used to detect the electrical activity of human brains exposed to physical products or images. A dataset, produced using the recorded voltage fluctuations resulting from ionic current within the neurons of the brain was then used to train a Recurrent Neural Network (RNN) algorithm to match them to the desirability of the design features. A GAN model conditioned by such design features was then trained to enable producing product concept images from EEG signals solely. This method is also important from the point of view that it allows AI methods to learn design requirements that cannot be explained verbally.

Gu et al. (2022) uses both CNN and DCGAN architectures to produce unique 3D spring concept models as shown in Fig. 9. Authors used perspective views (2D) of 130 3D spring models to train the DCGAN model to produce new 2D spring models and the CNN model to match the geometric and structural properties of the 3D models from the 2D spring images. A unique 3D spring model shown in the figure was then created according to the estimated geometrical and structural properties (Fig. 9). Producing this result with relatively small data is impressive, and the results can even be further improved by including the produced models in the training set.

5.3. Evaluation of ideas and concepts

A large number of design ideas and concepts generated in the early design stages should be evaluated to find the ideal design (Börklü et al., 2018). Features of design concepts such as creativity, innovation, applicability, diversity, aesthetics, and functionality can be evaluated by various methods (Oman et al., 2013). Evaluation based on these features often requires users and domain experts. However, user demands often contain uncertainties that need to be clarified. Furthermore, this conventional evaluation process becomes extremely difficult and time-consuming, especially for a large number of design options which is usually the case (Wu et al., 2020). With AI methods, an objective and rapid evaluation can be made, and thus significant time and cost savings can be achieved (Romeo et al., 2020).

Several research works have been performed to evaluate the quality and success criteria of design options depending on quantitative and qualitative variables using AI methods. Both variables can be evaluated with fuzzy set theory. In one application of this approach by

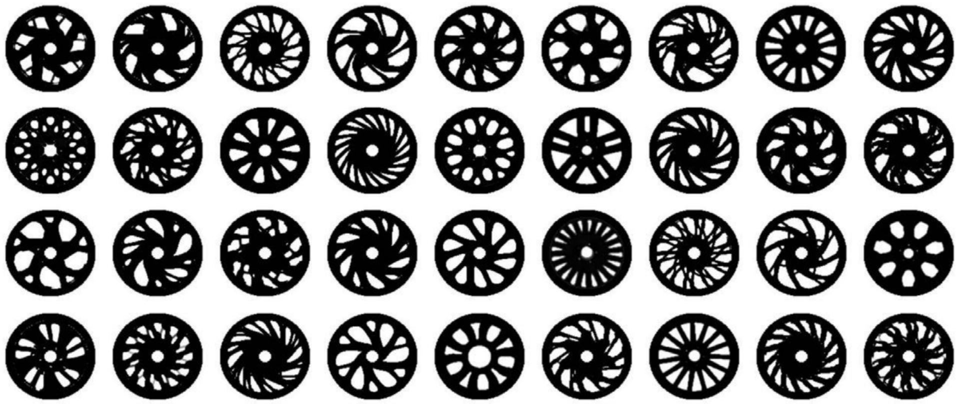

Fig. 8. 2D Wheel rim concepts produced with the BeGAN model (Oh et al., 2019).

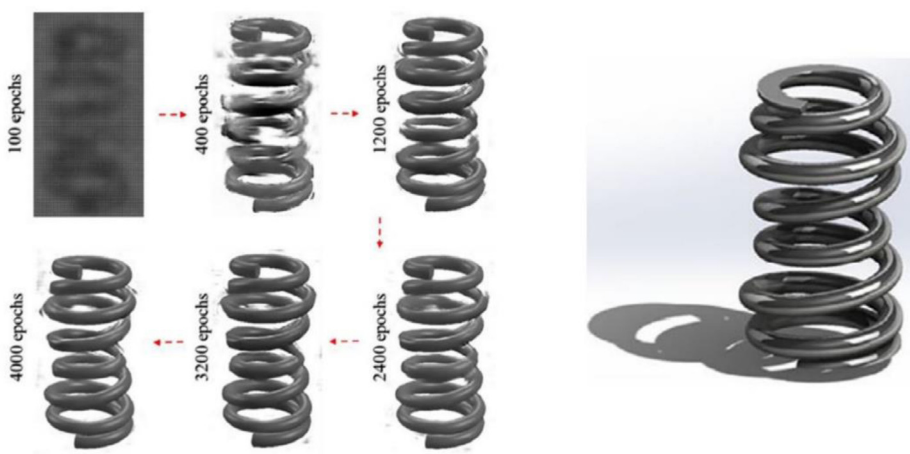

Fig. 9. Generating novel 3D concept models from 2D images by combining CNN and DCGAN models (a) 2D spring images produced with the DCGAN model (b) A 3D spring model created with geometric parameters estimated by the CNN model (Gu et al., 2022).

Sharif Ullah (2005), chair conceptual designs were evaluated using the fuzzy logic approach. The best design option was determined based on aesthetics and functionality criteria. Many other researchers also used fuzzy logic or a combination of this algorithm with other methods. For example, Ayağ (2005) combined Analytical Hierarchical Process (AHP) and fuzzy logic method to eliminate poor design options in the early design phase. Sarfaraz Khabbaz et al. (2009) suggested using fuzzy logic as well as a mathematical calculation for material selection in engineering design. Ko (2010b) developed a fuzzy elimination and evaluation method that takes aesthetic parameters into account in conceptual design. Yüksel and Börklü (2021) combined the fuzzy logic method with Pahl and Beitz's conceptual design for fast and precise concept evaluation in the early design phase. The validity of the method is demonstrated for a wheelchair conceptual design. In another rule-based AI method, Wang (2011) developed a hybrid Kansei engineering design expert system to determine the relationship between product form elements and product images. This method can also be used together with grey system theory to determine and evaluate customer expectation levels.

Machine learning algorithms have also been used to evaluate ideas and concepts. Toh et al. (2017) suggested using this method with data mining to determine the creativity level of ideas. Here, random forests machine learning models are trained with a small number of labelled data to detect creative ideas with 90% accuracy. Li et al. also used machine learning to determine the relationship between product images and customer emotional responses (Li et al., 2018). Product information data was automatically obtained from online shopping sites with a data mining application (Web Spider). The data were then used in an online evaluation questionnaire to obtain labelled datasets. Design options were evaluated and compared with four different machine learning methods (Support vector regression, Ridge regression, Classification and regression tree, and Artificial neural network). Design options were evaluated and compared with four different machine learning methods (Support vector regression, Ridge regression, Classification and regression tree, and artificial neural network). Fig. 10 shows the use of an artificial neural network, the most common of these methods, to understand the relationship between design parameters and customer expectations. The input layer is the basic feature of the product. The output layer expresses how the product is perceived by the customer.

Other machine learning algorithms such as Naive Bayes and Artificial neural network have also been used to evaluate design concepts in large solution space. Hao et al. (2019a,b) determined the innovation values of the design concepts with the artificial neural network. The applicability of these design concepts was then evaluated using the Naive Bayes algorithm. The authors aimed to narrow the wide solution space to highlight promising solutions in this work. This work showed that the degree of innovation and applicability of the design concept can be predicted with machine learning algorithms with a high success rate. However, manual creation of the knowledge base (created by examining the products obtained from popular e-commerce sites) prolongs the process. Camburn et al. (2020) studied text classification from crowdsourcing. Large databases of ideas obtained were scanned with an unsupervised neural network, and success levels were evaluated. Wang and Liu (2021) trained a multi-stage artificial neural network for the use of large volumes of data (design, production, use, etc.) produced in the Product Life Cycle (PLC) for the concept product design evaluation process. Particle Swarm Optimization and Adam (PSO-Adam)

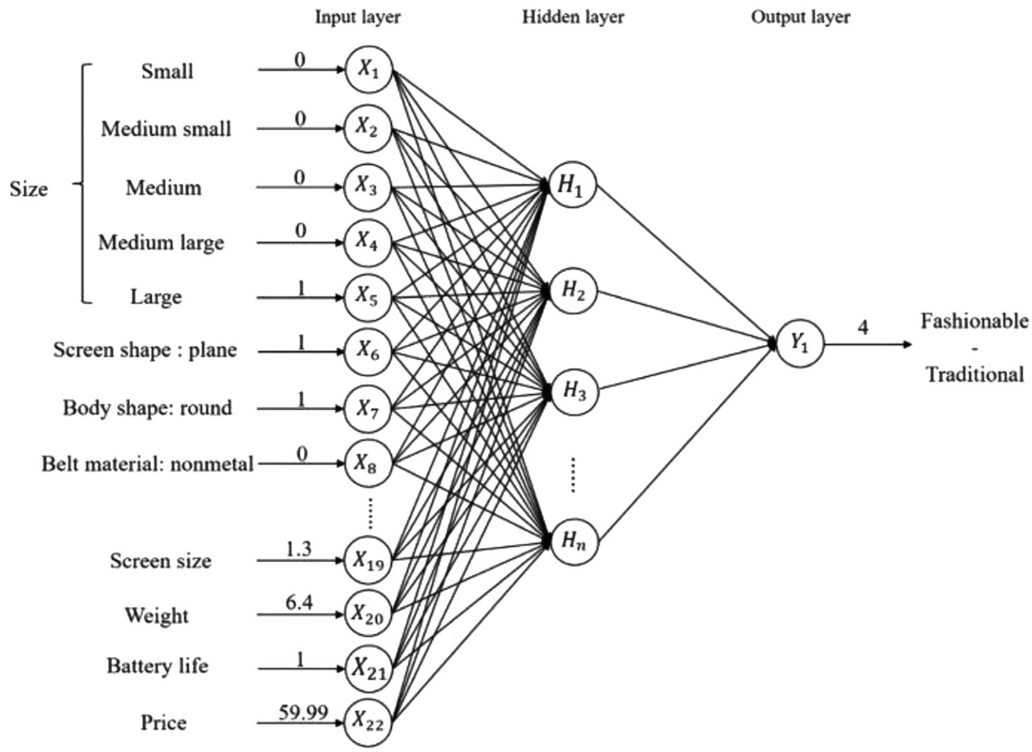

Fig. 10. Artificial neural network model used to understand the relationship between design parameters and customer expectations. The input layer is the basic feature of the product. The output layer expresses how the product is perceived by the customer. The network model was created for a smartwatch design (Li et al., 2018).

algorithms were also included in the training process. While PSO is used to determine the initial parameters required for network training, the Adam algorithm increases the ability to adapt to changing network parameters. Authors claim to increase the speed and accuracy of design evaluation processes with this combination method.

Deep learning algorithms can also be used to reveal and evaluate ideas at the desired level of creativity. Han et al. (2020a,b,c) proposed a data-driven approach to generate innovative ideas and evaluate the conceptual distances of ideas. They determined the level of creativity by measuring the distance between the concepts. As the distance between the two concepts increases, the level of creativity increases, and the level of applicability decreases. Two different tools, ConceptNet and TechNet, were used for distance measurement. Similar work was also undertaken by Chen et al. (2019) using a semantic network algorithm to calculate ideal concept distances and generate creative ideas. Generative design methods are also widely used, and successful results are obtained (Gatys et al., 2016; Yu et al., 2019a). In one example by Shu et al. (2020), a functional generative design model was developed to evaluate design concepts. Here, 3D airplane models produced by GANs, a generative modelling algorithm, are evaluated by Computational Fluid Dynamics (CFD) analysis. High-performance aircraft models with the lowest drag force are included in the training dataset for retraining the GANs model in Fig. 11. Additional high-performance 3D models in the initial dataset as verified by the CFD analysis were then used to increase the GAN model success. As the number of training increases, the probability of producing high-performance aircraft models also increases. Hence, ideal concepts with the lowest drag force performance can be obtained by repeating this concept generation and evaluation cycle. This process allows generating functional concepts unlike previous works focusing only on aesthetically appealing designs. Furthermore, the presented approach provides faster convergence with some compromise in precision as compared to the topology optimization method. In another concept evaluation study using a data-driven AI approach, Wu et al. (2020) suggested using CNN models to evaluate design concepts from product posters created for design competitions. The model is trained with seven different CNN architectures that categorize the product posters as awarded, pre-selected, and eliminated designs. The Squeeze and Excitation-Focal Loss-Residual Network (SEFL-ResNet) model, one of the CNN architectures, used in this work achieved 70,79% classification success. However, this approach considers the aesthetic quality only and ignores important other factors such as applicability, manufacturability, and cost making it insufficient in terms of evaluating the design quality. Nevertheless, it may be useful in automatic pre-assessment processes. In another study, Yoo and Kang (2021) used an explainable artificial intelligence method based on deep learning (i.e. CNN) to predict the machinability of complex structures in the CNC machining process and the costs involved during the conceptual design phase. Model visualization with 3D Gradient-weight Class Activation Mapping (Grad-CAM) was used to depict the machinability and estimated production costs. Regional machining difficulties are indicated on the model suggesting locations where a design modification is required. The method here is generally based on data collection, processing, and visualization steps to explain the prediction results.

5.4. Design optimization

Design optimization is used to determine the ideal product configuration, improve product features, increase functionality, reduce costs, produce lighter parts, etc. Here, it is generally aimed to create/determine the best design under various constraints. In the traditional optimization process, a mathematical formulation of the problem is created with advanced engineering knowledge to determine the optimal design variables. Recently, artificial intelligence methods such as evolutionary algorithms, swarm intelligence, artificial neural networks and machine learning have been increasingly used by designers to solve design optimization problems with minimal engineering knowledge.

The swarm intelligence algorithms are inspired by the collective behaviour of social creatures from insect colonies to human societies. They can find optimal solutions through cooperation, interaction, and

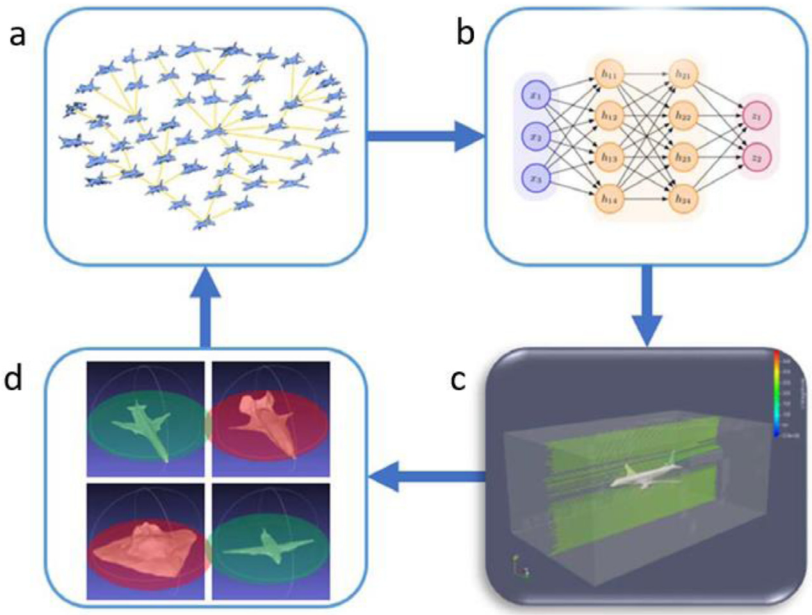

Fig. 11. The functional generative design model by Shu et al. (2020); (a) The design dataset (b) New concepts generated with the GAN model (c) CFD analysis of the generated concepts (d) High-performing concepts are verified by CFD included in the training dataset, and the process is repeated.

competition. Numerous algorithms and their variations are developed in the optimization such as Particle Swarm Optimization (PSO), ant Colony Optimization (ACO), Differential Evolution (DE), etc. (Johnvictor et al., 2022; Skinner and Zare-Behtash, 2018). Particle Swarm and Ant Colony Optimizations (PSO and ACO) have become some of the most preferred heuristic approaches for optimization problems due to their low memory requirement and ease of application. PSO is inspired by the social behaviour of bird flocking or fish schooling, while ACO imitates the foraging behaviour of real-life ants (Shelokar et al., 2007). PSO yields high performance giving solutions closer to the optimal for engineering problems (Engelbrecht, 2007). Zhengtong et al. (2019) presented PSO algorithms for 15, 39, and 47-planar truss problems with multi-material layout and design configurations. They declared that PSO can be applied to more complex optimization problems of truss and frame structures. Jafari et al. (2021) integrated the Cultural Algorithm (CA — a branch of evolutionary computation) into PSO to create a hybrid algorithm with increased efficiency for the optimal design of truss structures. The efficiency of PSO is also increased by integrating reinforcement learning algorithms (Samma et al., 2020). Samma et al. (2016) evaluated this hybrid algorithm using several benchmark problems such as the gear design and the pressure vessel design problems. The experimental results show that Reinforcement Learning-based Particle Swarm Optimization is useful, and it outperforms a number of state-of-the-art PSO-based algorithms. Design optimization problems may have a highly non-convex behaviour that can be handled by a combination of PSO and ACO algorithms, named the Particle Swarm Ant Colony Optimization (PSACO).

Differential Evolution (DE) optimizes a problem by iteratively trying to improve a candidate solution with regard to a given measure of quality. DE can be used for multidimensional real-valued functions without using the gradient of the problem being optimized as is required by classic optimization methods. It can also be used on discontinuous, noisy, change over time, etc. optimization problems (Talbi, 2009). DE optimizes a problem by preserving a population and creating new candidate solutions by combining existing ones according to its simple formulae and then keeping the candidate that has the best fitness value on the optimization problem. In this way, the optimization problem is treated as a black box that does not need the gradient. Nadimi-Shahraki et al. (2020) proposed a Differential Evolution (DE) algorithm and evaluated it on a benchmark suite that includes unimodal, multimodal, hybrid, and composition functions and four complex engineering design problems. In addition to these heuristic optimizations, it can specify wing morphing design parameters that provide enhanced flight performance and expanded flight envelopes. Leylek et al. (2010), considered and optimized the design of bat wings using Simulated Annealing (SA) that is proven robust and reliable in finding global optima as the optimization method (Corana et al., 1987) is presented in Fig. 12.

Genetic algorithms and mixed models are also widely used in the creation and selection of the ideal design variants, especially in large solution spaces (Chen and Martinez, 2012; Matei, 2015). Jiao et al. (2007) tried to determine the optimal product family configuration with the Generic Genetic Algorithm (Dongxu et al., 2020). The aim here was to increase product variety with the same design cost. Common product components such as parts and subsystem interface connections in the product family are optimized to reduce design cost and production process lead time. A general coding scheme has been created in this work to adapt it to various product family designs. This work demonstrated viable solutions for part and component integration to meet customer expectations. In another work, Brintrup et al. (2007) presented the Multi-Objective Interactive Genetic Algorithm framework that addresses quantitative and qualitative criteria together. The validity and applicability of the method were tested in the factory layout problem. It must be noted here that the interactive genetic algorithm has some disadvantages such as user fatigue and excessive population noise which may jeopardize its overall success. Dou et al. (2016) developed the "Multi-Stage Interactive Genetic Algorithm (MS-IGA)" to overcome such drawbacks. Here, the design problem is divided into meaningful parts which are evaluated step by step. This way, the system actually approaches a real evolution model. The aim here is to increase the customer-product interaction partitioning the design problem from simple to complex stages. The proposed framework and other interactive genetic algorithms (i.e. Traditional IGA and MAN-IGA) have been applied to a car dashboard design to compare the success and efficiency. The authors state that the algorithm can reach accurate results faster than the other methods. Yildiz (2013) proposed a mixed optimization technique based on the Differential Evolution algorithm and Taguchi method. The Taguchi method here is used to generate a strong initial population enabling to increase the convergence rate in finding the global optimum. The presented method has been compared with methods such as genetic, artificial bee colony, and differential

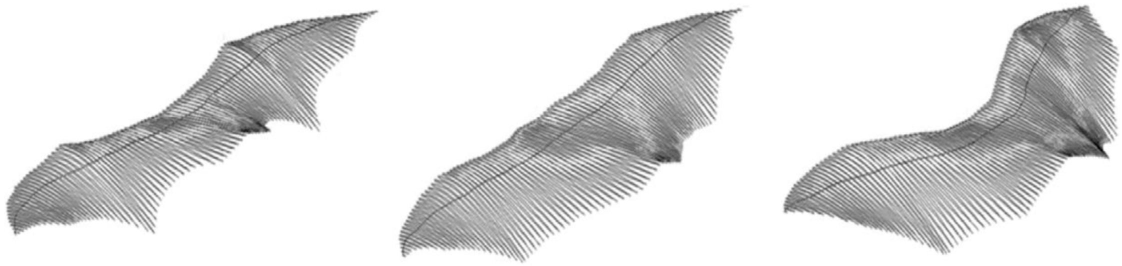

Fig. 12. Different bat-wing topologies for wing morphology optimization using simulated annealing (Leylek et al., 2010).

evolution algorithms. In tests on a vehicle component, it has been stated that this hybrid technique outperforms other optimization methods.

Zha (2004) proposed the Hybrid Dual Cross-Mapping Neural Network (HDC MNN) model to identify ideal product configurations under various constraints. The model was formed by a combination of Back-Propagation Neural Network (BPNN) and Hopfield Neural Network (HNN). Here, the BPNN identifies and classifies the net weight values, whilst the HNN uses these weight values to optimize the design. Lin et al. (2009) also used artificial neural networks to understand (interpret) the relationship between product image and form features. The goal here was to determine the ideal product form and the optimum product configuration. With this model, simple product form designs can be created. Chen and Chiang (2010) used the back propagation neural network to determine the optimum design configuration from consumer satisfaction data. The AI model is trained with the information obtained from surveys as well as purchasing data. The model was then used to generate the optimum product configuration with the highest customer satisfaction. Similarly, to this work, Kutschenreiter-Praszkiewicz (2013) used the ANN and QFD method to determine product features suitable for customer expectations and improve product quality and create value. Another interesting work by Nguyen (2015) used the Repetitively Enhanced Neural Network (RENN) method for the optimization of the unmanned aerial vehicle wing. RENN are neural networks that avoid over-fitting created by the development of traditional neural networks. In this method, back propagation algorithm and Bayesian learning algorithm are applied together to avoid overfitting. The model was trained with geometrical constraints to optimize the maximum lift coefficient, angle of attack, drag coefficient, and thickness ratios. Fig. 13 shows optimum blade shapes according to RENN and simulation analysis. Although there was little geometrical difference between these shapes, RENN provided an 81.2% reduction in computation time compared to simulation analysis (i.e. CFD). In a more recent work, Zhang et al. (2020a,b) proposed a new optimization framework by combining Neural Network Algorithm (NNA) with high performance in finding global optimum and Teaching-Learning-Based Optimization (TLBO) methods with a high convergence rate. This combination model allowed finding the global optimum quickly.

Machine learning algorithms other than ANN were also used for various design optimization studies. For example, Sharpe et al. (2019) compared four different machine learning algorithms for this purpose. The advantages and disadvantages of Random Forrest, Gaussian Naive Bayes, Neural Network, and Support Vector Machine algorithms in discovery and optimization problems are investigated. These algorithms were compared in six different design problems and as a result, no obvious superior technique over others could be determined. Appropriate machine learning algorithms may differ depending on the problem characteristics. Regardless of the method used, the use of machine learning algorithms in optimization problem solutions gives designers an intuitive view making the overall design process more efficient.

Recently, deep learning methods have been increasingly used especially for topology and shape optimization problems (Burnap et al., 2016; Umetani, 2017; Yu et al., 2019b). Although the applicability of the design is determined by its form, function, and behaviour characteristics, generative design models based on deep learning algorithms generally aim for form improvement. Modern generative model studies, on the other hand, aim to develop form and function together with physics-based simulation analysis. One example of this is the aforementioned study by Oh et al. (2019) where aesthetic designs with high engineering performance were demonstrated with topology optimization and the integration of Boundary equilibrium GANs (BeGANs). It must be noted that a large amount of data is needed to avoid over-fitting errors in generative models. To satisfy this critical prerequirement, the data obtained from topology optimization is used as training data in this study, and high-performance models are produced. An example is given in Fig. 14 where wheel concepts are produced with BeGAN in terms of compliance, innovation, and cost. Shu et al. (2020) carried out a similar study on 3D aircraft models. Drag forces were determined by CFD analysis of the 3D aircraft models produced with GANs. Aircraft models with low drag coefficients are added to the initial dataset to improve the performance of GANs. However, a large number of iterations may be required to approach the optimum drag coefficient values of the final models. Also, the performance of this method is lower as compared to topology optimization. Despite this, the method may still provide valuable results with lower cost and minimal design and engineering domain knowledge compared to topology optimization. In another study, Sasaki et al. (2021) applied topology optimization to an Internal Permanent Magnet (IPM) motor with high torque performance using motor cross-section images with a combination of genetic algorithm and supervised deep learning (CNN) based XAI. Gradient-weighted Class Activation Mapping (Grad-CAM) was used to understand which parameters were more effective in reaching the result of this Deep Neural Network (DNN). Authors claim that both high performance and interpretable topology optimization have been achieved with explainable artificial intelligence.

Although the AI optimization algorithms mentioned so far are successful for many design optimization cases, they are inefficient for complex and large systems with huge design variables and alternatives because this leads to a prolongation of the design processes and a hence increase in costs (Du and Zhu, 2018). In order to cope with the problems caused by high dimensionality from different perspectives, different strategies/techniques such as design domain reduction, critical parameters identification, parallel computing, increasing computer power, decomposing design problems into sub-problems, mapping, and visualizing the design space have been developed (Shan and Wang, 2010; Shi et al., 2013; Zheng et al., 2015; Wu and Gary Wang, 2019). Using these strategies alone or in combination, high-dimensional design problems can be overcome. For example; Du and Zhu (2018) proposed a new data-driven methodology (i.e. statistical learning) to implement the design domain reduction and critical parameters identification strategies simultaneously for complex mechanical system optimizations. Critical design parameters and parameter value ranges were determined using a decision tree statistical learning algorithm. The applicability of the method was tested on an automobile S-shaped beam and compared with conventional methods. Another critical design parameter determination study was used in blade design, which is one of the multidimensional engineering optimization problems. Echeverria

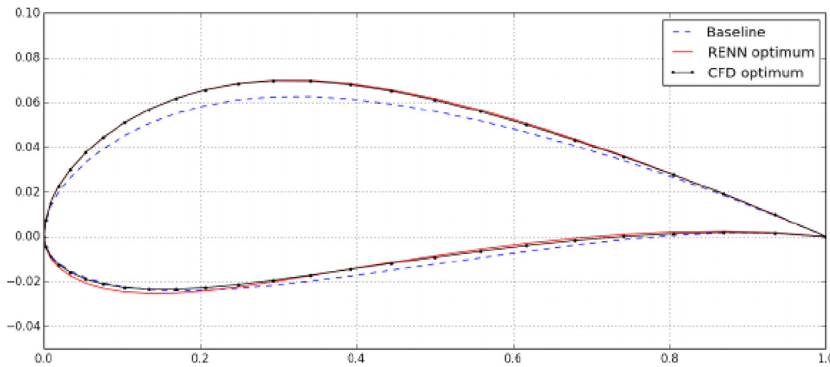

Fig. 13. Aerofoil shape optimization results by CFD analysis and RENN method. Blue dashed line baseline aerofoil, red line RENN optimum aerofoil, and black dotted line CFD optimum aerofoil (Nguyen, 2015).

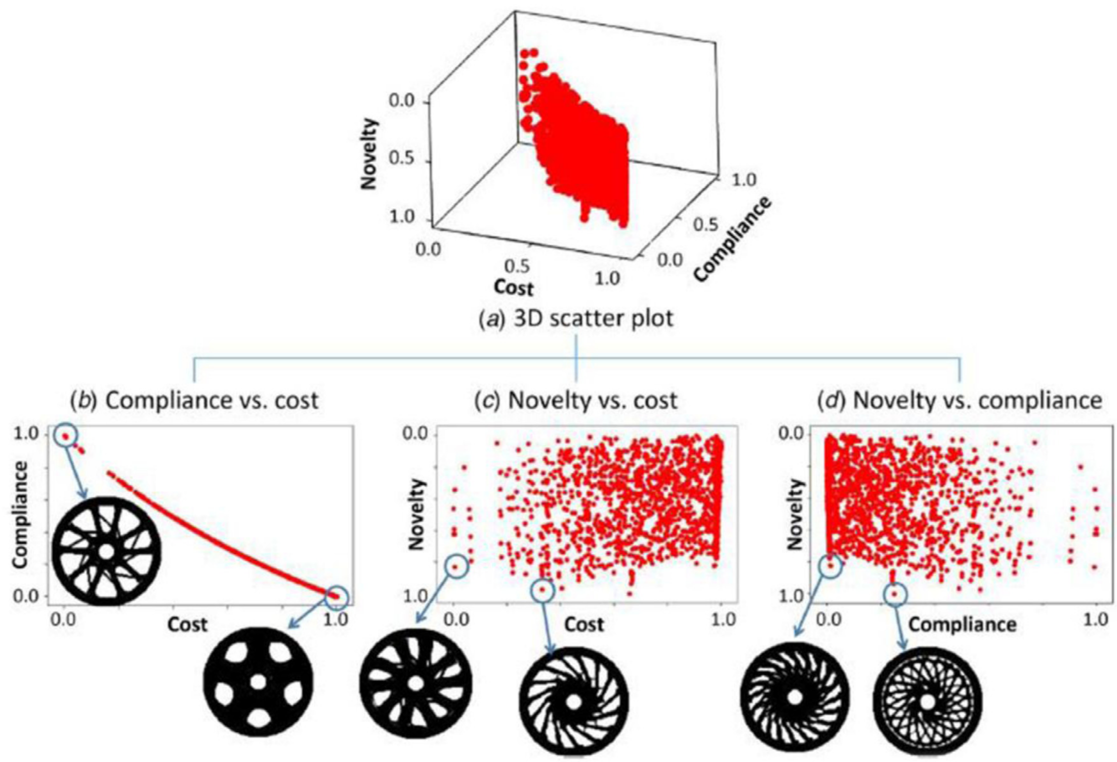

Fig. 14. Analysis of wheel rim concepts produced with BeGAN in terms of compliance, innovation and cost (Oh et al., 2019).

et al. (2018) identified design parameters that lead to undesired dynamic load amplification before optimization with statistical learning methods such as logistic regression and decision trees. Apart from these strategies, there are two different optimization strategies for solving high-dimensional, computationally expensive, black box (HEB) problems: model approach-based techniques and coarse–fine model-based techniques. The Model approach-based optimization techniques use the information obtained by optimizing the computationally cheap model to optimize the expensive one. In coarse–fine model-based techniques, on the other hand, a coarse model completes the optimization in a short time and at a low cost, while the task of the fine model is to test the accuracy of the optimization (Shan and Wang, 2010; Wang and Shan, 2006).

5.5. Design support systems

The use of AI methods in design can be either directly or indirectly implemented to provide full or partial support to the design process and assist designers in various decision-making stages (Fu and Zhou, 2020). Ideas and concepts generated by AI may not always end up with a good design. However, it can be brought to the desired level gradually with human intervention (Liao et al., 2020). Of the all AI methods, expert systems, machine learning, and deep learning algorithms are the most widely used decision support tools to assist designers (Bickel et al., 2019; Kaljun and Dolsak, 2011; Wang, 2011). Zarandi et al. (2011) developed an expert system to support the end-of-life engineering approach for material selection in sustainable product design. Material selection is an important parameter that determines the environmental impact of the product and is fairly difficult to manage in the early stages of design due to a lack of knowledge. In addition, performing end-of-life engineering for each material option is a long and costly process. The expert system developed by Zarandi et al. (2011) first evaluated all the materials and then removed the non-applicable ones from the list. This hence allowed designers to choose from a smaller number of material options.

Machine learning is another widely used algorithm to assist designers and technical personnel in determining which option is more

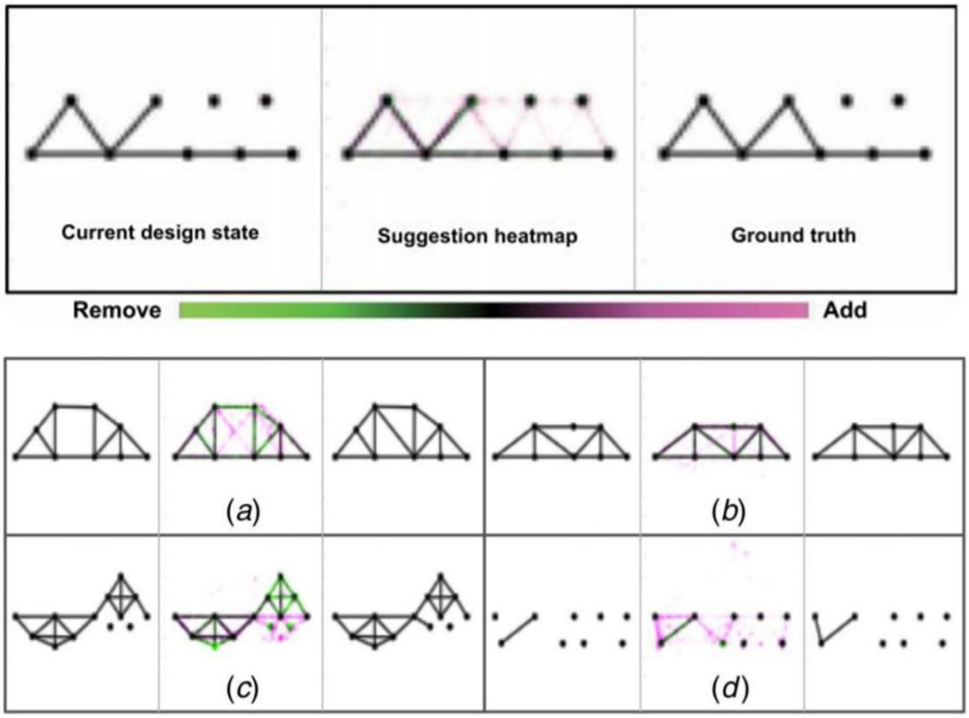

Fig. 15. Simple truss systems with the suggested design steps of the deep learning model trained with sequential design visuals the green (Alanis et al.). From left to right; the current state of the design, the proposal of the deep learning model, and the next actual design step. The green colour indicates the areas where the material will be removed, while the pink colour indicates the areas where the material will be added (Raina et al., 2019).

appropriate during such design decision problems (Romeo et al., 2020; Sarica et al., 2020). Romeo et al. (2020) proposed an innovative cloud-based Design Support System (DesSS) to predict the machine structure and material from operational data (e.g. torque, efficiency, speed, cost, etc.) obtained in the production and/or process simulations and vice versa. The ultimate goal here is to guide designers to estimate important performance parameters for an ideal design solution. The model accuracy was presented with two different datasets and three different machine learning algorithms (Decision/Regression Tree, k-Nearest Neighbours, and Neighbourhood Component Features Selection). Akbari et al. (2020) also used a subcategory of a machine learning algorithm (i.e. Recursive Neural Tensor Network — RNTN) as a decision support system for an innovative product development process taking customer comments into account. Customer opinions collected from social media platforms were analysed to determine their emotional state toward a particular product. Unlike other customer sentiment determination studies, this algorithm was designed to examine how a user's comment affects other users. The information obtained as a result of the mood determination study was presented to the designer to be used in the innovative product development process. In addition, the cost necessary to create a positive opinion on the community was also calculated. Ferrero et al. (2020) used machine learning algorithms to facilitate function-based design studies in decision-supporting context. The functions that each product component can fulfil are determined and classified with the classification and regression trees (Saxby et al., 2020) as well as Classification Based on Association Rules (CBA). The data required for the study was obtained from The Oregon State Design Repository (OSDR), which was established to support engineering and design activities. Function-based solutions here were filtered, allowing designers to work in a narrower space. This method can assist especially novice designers in selecting product components suitable for the function.

Raina et al. (2019) used a deep learning-based design support system for an engineering design problem following the mental processes behind the consecutive design stages. The effectiveness and applicability of the method are demonstrated in a lattice design problem in Fig. 15. All stages of the design process were recorded with screenshots, and deep learning models (i.e. Auto-encoder and CNN) were trained with sequential images of the design process. The model detected the current state of the design and made a design proposal for the next step. Simple truss designs produced with deep learning support following the unknown mental activities behind the design process were evaluated according to the factor of safety and strength-to-weight ratio. Since AI algorithms are known to lag behind human designers in matters such as abstract thinking, creativity, and explainability, this method of developing design strategies following the initially unknown human mental process is especially valuable in creating design strategies based on human designers' knowledge and experience.

5.6. Modelling and retrieval

Modelling in this context is defined as creating 3D images of the determined design concept and its geometric representation in a computer environment. It is a mathematical expression of the volume occupied by the object in space. It is a valuable asset for the designers as well as the product stakeholders, giving an idea of the general structure of the product. 3D models can be created after the conceptual design phase, which takes weeks or months in the traditional engineering process. In addition, a limited number of 3D models can be created depending on human capacity. AI algorithms/methods can assist designers in automating the modelling process, executed by designers in a CAD environments (Bryant et al., 2008). The development of 2D and 3D model repositories in recent years has facilitated the automation of 3D model generation via AI (Newstetter and Svinicki, 2015).

A fuzzy logic integrated tool was proposed by Gologlu and Mizrak (2011) to generate 3D CAD models based on customer demands. Verbal and ambiguous demands from customers were transformed into product parameters with the fuzzy logic algorithm. The parameters were then used to generate a solid model of the product automatically in the CAD environment. Hence, customers could be directly involved in the design process converting their requests into CAD models without the need for modelling knowledge.

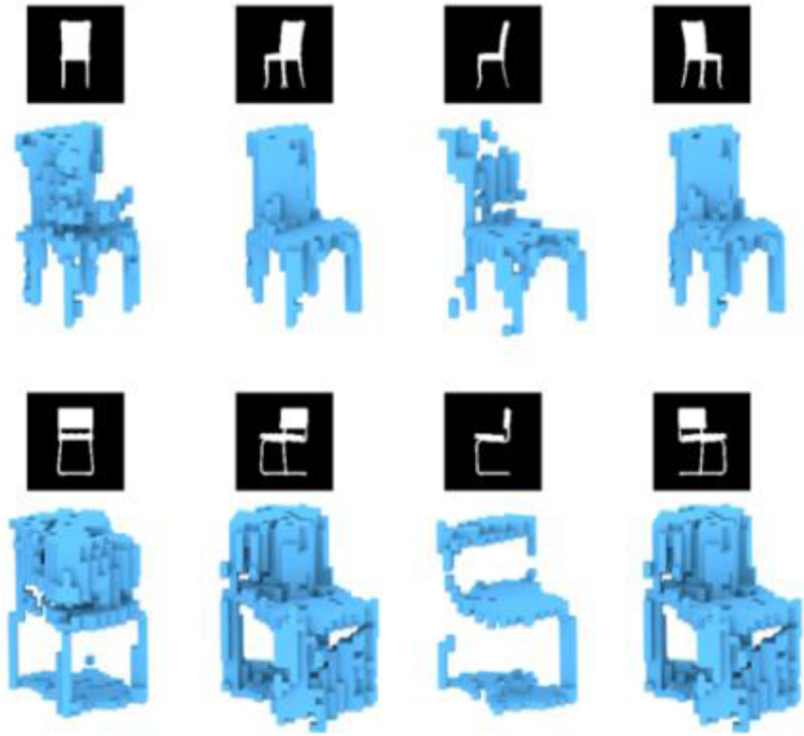

Fig. 16. 3D chair models produced with PrGAN model (Gadelha et al., 2017). Chair models in each row are generated with 2D images taken from different angles.

Deep learning methods are especially useful in 3D CAD model generation as they can provide automation in generating unique 3D model variations or even 2D to 3D model transition (Dosovitskiy et al., 2016; Nobari et al., 2021; Saxby et al., 2020; Shu et al., 2020). However, the latter concept, creating 3D models from 2D images, is a more complex process (Sun et al., 2018; Wu et al., 2017; Zhou et al., 2021). Wu et al. (2016) developed deep learning algorithms to produce realistic 3D visuals and also generated 3D models from 2D visuals using 3D-GAN and 3D-VAE-GAN (Variational Auto-Encoder and GAN) models respectively. The latter model consisting of the encoder, decoder, and the discriminator was trained with both 2D images and the corresponding 3D models so that a 3D model could be created from a single 2D image. Gadelha et al. (2017) obtained 3D structures from 2D product images with the Projective Generative Adversarial Network (PrGAN). By adding a projection module to the generative network, 3D models of 2D images given from arbitrary angles were obtained. The study showed that PrGAN trained with 2D visuals performed better than the GAN trained with 3D models. However, the PrGAN results are relatively low-resolution models since the models are represented by voxels as seen in Fig. 16. Furthermore, the lack of shadows and missing interior details can cause incorrect geometry. Sun et al. (2018) developed a dataset and method to generate 3D models from 2D visuals using a GAN algorithm. The data set, called Pix3D, consists of paired image models in nine different categories. Reference points were aligned to create the visual and 3D model relationship. Weak and insufficient visual-3D models were eliminated to improve the quality of the dataset. The developed model generating 3D models from 2D images was compared with alternatives trained on the Pix3D dataset (i.e. to 3D-VAE-GAN Wu et al., 2016, MarrNet Wu et al., 2017) and proved to perform better. Zhang et al. (2020a,b) developed CNN-based FilterNet and RankNet models for 3D CAD model retrieval corresponding to given hand drawing or technical drawing views. As the name suggests, the FilterNet model filters categories in the library. On the other hand, the RankNet model calculates the similarity between the input variables and CAD models. This process allows reusing library 3D CAD models and hence reduces design time and costs.

6. Discussion

Previous sections provided a comprehensive critical review of current AI methods used in engineered product designs. Classical AI methods (i.e. fuzzy logic, genetic algorithm, and neural networks) and modern data-driven approaches (i.e. machine learning and deep learning) are discussed to identify suitable AI methods for design processes/problems and assess strengths and weaknesses. Comprehensive information and criticisms given on this subject can be a guide, especially for new researchers and designers.

Fig. 17 shows the number of designs included in AI articles data derived from the Scopus database between the years 2006 and 2021. As evident from the figure, over the past decades, there has been a significant increase in the number of articles involving AI methods in design studies. The developments in high-performance computer machines have an important role in this dramatic increase (Coe, 2020). The increase in the number of articles is remarkable, especially after 2014.

Fig. 18 shows the usage rates of AI methods in design applications by year intervals. A clear shift in research intensity from knowledge-based AI methods to data-driven approaches is seen in this figure; machine learning and deep learning have been increasingly used in research studies and publications since 2015. Looking at the studies conducted between 2006 and 2009, the popularity of expert systems and genetic algorithms is 15% higher than other methods as seen in Fig. 18. In the early 2010s, machine learning-based design studies started to take place in the literature. Machine and deep learning methods, which account for approximately 64% of all design work involving AI, have become one of the most popular research topics in 2016 and beyond.

AI research topics in design have also changed over time as shown in Fig. 19. This figure shows the distribution of publications involving AI research topics methods elaborated in Section 5 (i.e. generate inspiration, ideas and concepts, evaluate ideas and concepts, and optimize design and modelling) between the years 2006 and 2021. As shown in the figure, most of the AI research topics in design (71%) between

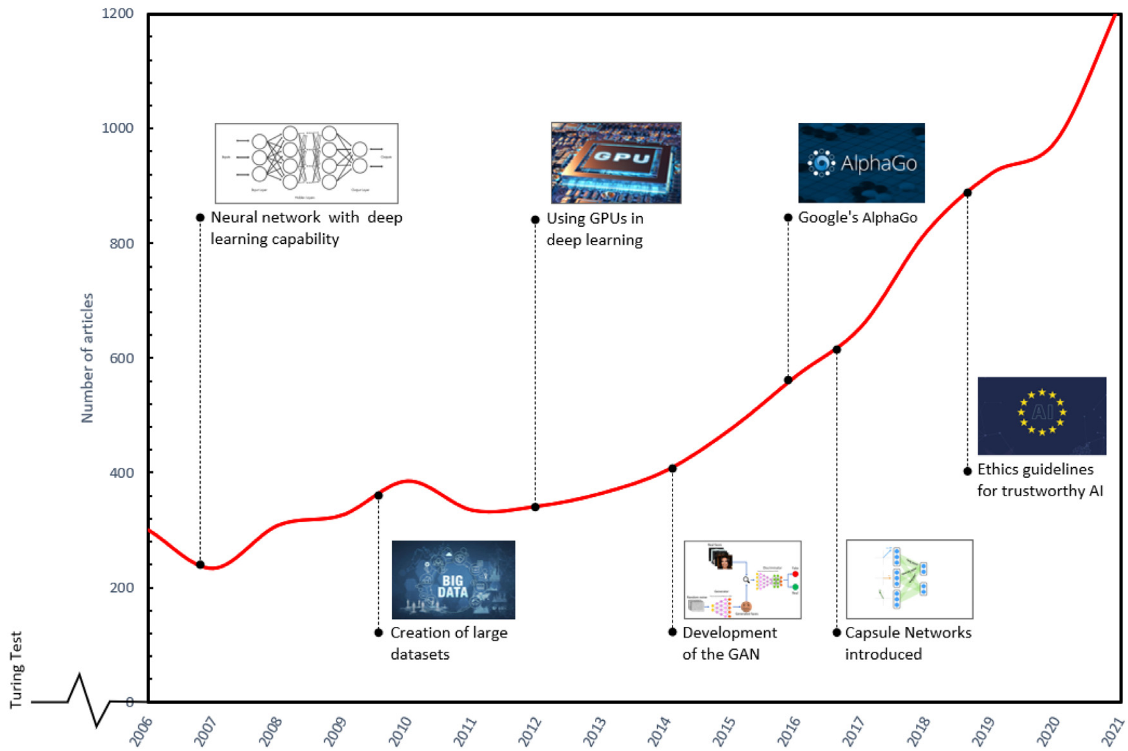

Fig. 17. The number of articles involving AI methods between 2006 and 2021 (Scopus).

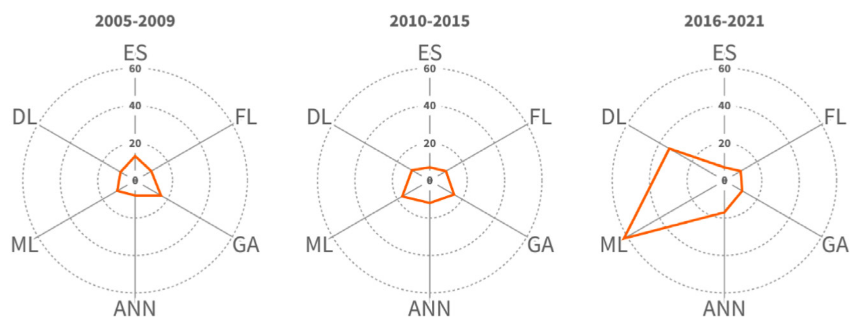

Fig. 18. Distribution of publications involving the design based on AI methods between 2006 and 2021, (ES: Expert Systems, FL: Fuzzy Logic, GA: Genetic Algorithm, ANN: Artificial Neural Networks, ML: Machine Learning, DL: Deep Learning).

2006 and 2009 were related to the Evaluation (E), Optimization (O), and Decision-Making (DM) processes. The next 5 years have seen an increase in optimization-based AI research and in the last 5 years, Ideation (Id) and Inspiration (In) studies become more pronounced and even the most dominant research topics with 37 percent (Fig. 19). An increase in Modelling studies (M) (generating 3D models mostly with deep learning methods) can also be realized in this time period.

The main purpose of using AI in design studies can be explained as easing the burden of the designer, reducing design costs, and accelerating processes and automation. Detailed information on AI applications is presented in Section 5. However, any AI method cannot be used in all design processes and required operations. The AI method needs to be appropriate to the nature of the problem. Commonly used AI methods and their most relevant applications in design studies with the most apparent pros and cons are summarized in Table 2. Generally, from the Table, data-driven design methods are free from subjective judgments and good to overcome phenomena such as psychological inertia that hinder the creativity of designers (Jansson and Smith, 1991; Nguyen and Zeng, 2017). Personal pre-judgments can be thrown into the background for the evaluation of ideas and concepts (Hao et al., 2019a,b; Su et al., 2020). Deep learning models that learn from data can quickly adapt to changing environment and conditions (Verganti et al., 2020) and requires minimal design and engineering knowledge. People-oriented fuzzy logic and expert systems, on the other hand, can work on the bases of rules that are created with human knowledge and experience and work with uncertain and incomplete data/information. The ability to operate with verbal expressions makes it possible to apply real-life problems. These methods can be used in the medical, defence, and financial sectors thanks to their high interpretability. In addition, the level of explainability can be increased by integrating fuzzy logic into some artificial intelligence methods (Bonanno et al., 2017; Garg, 2021). However, fuzzy logic models are limited out of evaluation, and decision-making processes are specific case-based problems and cannot be applied to any other problem. Gathering data/information, processing, and establishing the rule structure take time and a lot of effort. Genetic algorithms used in multidimensional design problem solutions are an optimization method that has been used for a long time. If the solution space is large and highly diverse which may increase the processing times, the probability of reaching the ideal solution increases. The model can be integrated with other methods to increase its performance (Dou et al., 2016; Majak et al., 2008; Yan et al., 2017). Many parameters that require experiences such as

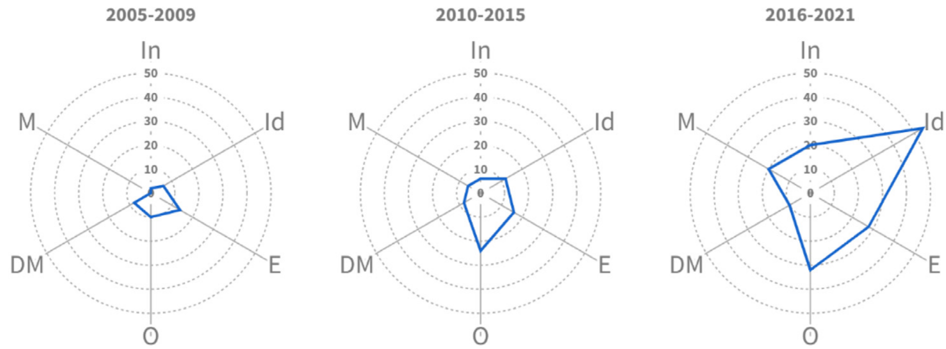

Fig. 19. Distribution of publications involving the topic of design based on AI methods between 2006 and 2021 (In: Inspiration, Id: Ideation, E: Evaluation, O: Optimization, DM: Decision-Making, M: Modelling).

Table 2
The advantages and disadvantages of application areas of artificial intelligence methods in design processes.

AI methods	Input variables	Design practises	Advantages	Disadvantages	Model Interpretability
Machine Learning	Dataset	Inspiration Idea generation Concept generation Evaluation Decision-making Design support Optimization	∘ There is little need for human intervention. ∘ It can be used in multidimensional design problems. ∘ Applicable to many designs' problems. ∘ It can perform well on small datasets. ∘ Computationally efficient ∘ Algorithms are suitable for supervised, unsupervised, and reinforcement learning.	∘ There are many machine learning algorithms. It is difficult to determine which algorithm to use. ∘ Creating a labelled dataset is difficult, Data pre-processing may be required. ∘ Interpretation of results requires expertise. ∘ High error rate. ∘ Training time depends on the hardware.	∘ Partial interpretability. ∘ Some simple ML algorithms such as linear regression, logistic regression, decision trees, K-nearest neighbour, naïve bayes can be interpreted/explained.
Artificial Neural Networks	Dataset	Idea generation Design support Optimization Decision-making	∘ It can work with incomplete and noisy information. ∘ It can be used in multidimensional design problems. ∘ It is suitable for supervised and unsupervised training.	∘ Training time depends on the hardware ∘ It cannot be explained exactly how the solution was reached. ∘ Only works with numeric data ∘ Feature selection requires experience.	∘ Low interpretability. ∘ It can be interpreted after model training with the help of simplification, feature relevance, local explanation, textual justification, and visualization techniques.
Deep Learning	Dataset	Inspiration Idea generation Concept generation Evaluation Decision-making Design support Optimization Modelling	∘ Requires minimal engineering and domain knowledge. ∘ The network decides for itself which features to learn from the dataset. ∘ Applicable to many designs' problems. ∘ Increases creativity. ∘ It adapts to changing situations. ∘ The higher the number of data, the higher the model performance. ∘ Algorithms are suitable for supervised, unsupervised, and reinforcement learning.	∘ Network training requires a lot of data ∘ Design datasets are limited. Especially 3D models are difficult to reach. ∘ Creating a labelled dataset is difficult, Data pre-processing may be required ∘ Training times are long (depending on hardware). ∘ Generative models can only generate new forms. Their functional properties are weak. ∘ It cannot be explained exactly how the solution was reached.	∘ Low interpretability. ∘ It can be interpreted after model training with the help of simplification, feature relevance, local explanation, textual justification, and visualization techniques.
Genetic Algorithm	Population	Optimization Concept Evaluation Idea generation	∘ It is suitable for working in a large solution space. ∘ Can be used in multi-parameter optimization. ∘ It is faster than other optimization algorithms.	∘ Experienced genetic algorithm designers are needed. Determining population size, convergence rate, fitness function, mutation rate, and crossover rate requires experience. ∘ There is always the possibility of not finding the ideal solution.	∘ Low interpretability.
Fuzzy Logic	Knowledge/ Rule base	Evaluation Decision-making	∘ It can work with uncertain and incomplete information. ∘ It can be applied to complex design problems. ∘ Easy to apply to design problems. ∘ It is suitable for automation.	∘ Specific to the problem, difficult to apply to other problems. ∘ Building a rule base takes knowledge and experience. ∘ It is people-oriented. ∘ It can't develop creativity.	∘ High Interpretability. It can help explain other AI methods.
Expert Systems	Knowledge/ Rule base	Design support Evaluation Decision-making	∘ It ensures that its expert knowledge and experience is always used. ∘ It prevents errors arising from human situations (prejudice, fatigue, illness, etc.). ∘ Explain how the solution was reached. ∘ The results are reliable. ∘ It is suitable for automation. ∘ It can be updated.	∘ Specific to the problem, difficult to apply to other problems. ∘ Field expertise is required. ∘ Building a rule base takes knowledge and experience. ∘ It is people-oriented. ∘ It cannot develop creativity. ∘ It is difficult to apply to complex problems.	∘ High interpretability. ∘ Every result obtained is explained in the human–computer interface.

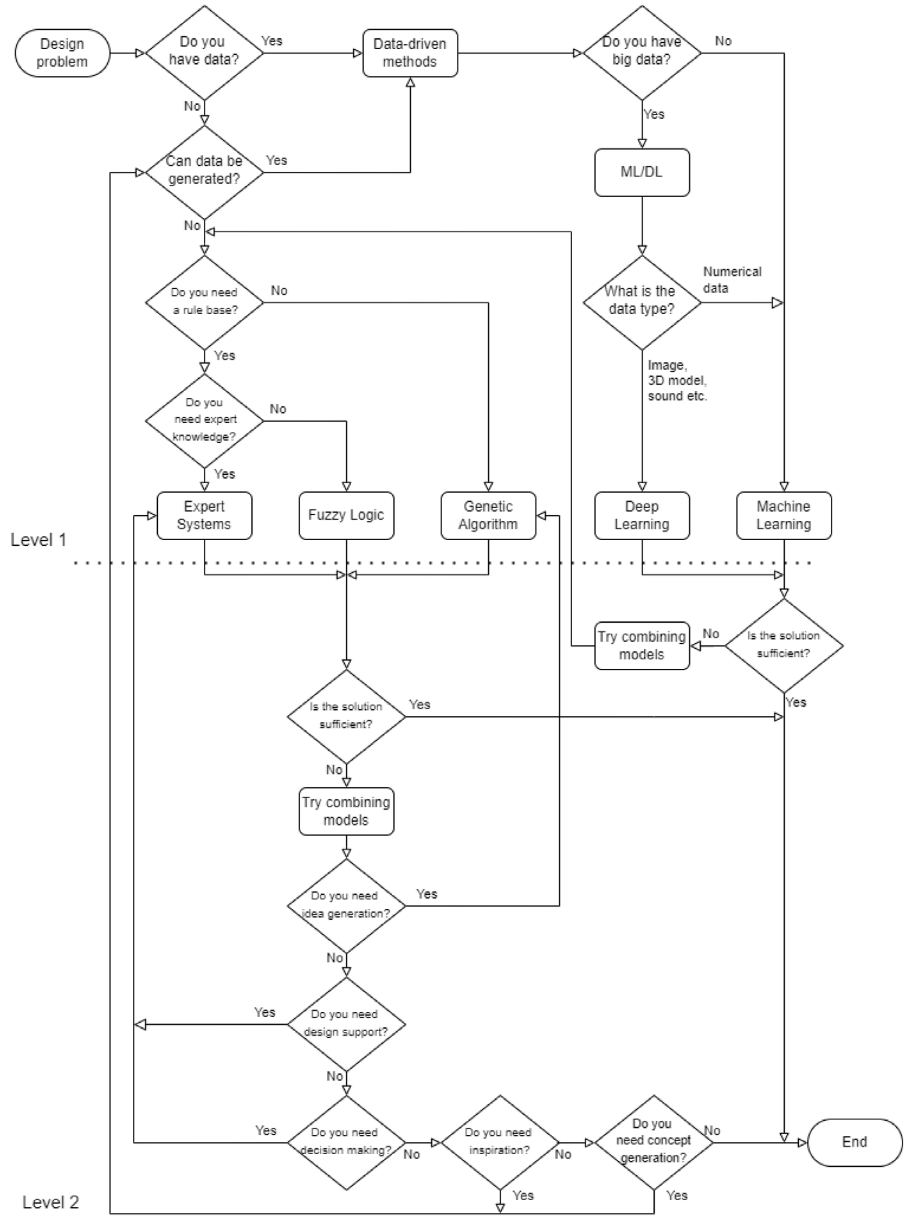

Fig. 20. The flow chart of selecting the proper AI method for design applications.

population size, mutation rate, crossover rate, and selection method affect reaching the global optimum. Apart from the genetic algorithm, there are many algorithms such as PSO, ACO, and DE, which are used to enhance the design performance (Johnvictor et al., 2022).

AI methods can be used in many applications depending on the requirements, availability, and availability of input parameters. First, fuzzy logic and expert systems from classical AI methods can assist designers in their evaluation and decision processes. Genetic algorithms and hybrid methods may be more effective in creating alternative solutions and determining the ideal design option. Data mining and machine learning methods, which are modern AI methods, yield efficient results in identifying customer demands, uncovering emotional needs, inspiring designers, generating ideas, making decisions, and optimizing. It is possible to transform sketches into pictures and create new images from 2D images, 3D models from 2D images, and new 3D models from 3D models with deep learning methods.

AI Algorithm Selection

Determining an appropriate AI method for the design problem provides many advantages in solving design problems. For this reason, it is extremely important to define the appropriate method for a given specific design problem which is the most complex part of the process. Especially designers who do not have sufficient knowledge about AI methods may have difficulty choosing the right method. The two-level AI method selection flow chart constructed to guide the designers based on successful applications reported in the open literature is presented in Fig. 20. The algorithm starts by asking the designer for the availability of data. This is because the availability of data to be used in solving the problem is one of the most important criteria in determining the right AI method. If the answer to this question is negative, then the algorithm asks if it can be generated as it may be possible to generate synthetic data in some cases (computational design, topology optimization, generative design, etc.). The presented algorithm suggests, one way or another, if sufficient data is available (or can be generated), data-dependent methods should be preferred. The appropriate AI technique is then selected by evaluating the number and types of data. On the other hand, either rule-based systems or genetic algorithms can be used in case of unavailable data. The criterion to choose one over the other is if there is a need for rule/knowledge bases that produce objective and somewhat creative results. A genetic algorithm is the only alternative if no rule base can be established. Otherwise, expert systems or fuzzy logic are preferred according to how the rule base information is obtained. The algorithm suggests an artificial intelligence method at the end of the first level which is suitable/useable if the designer's problem matches one of the common implementations of this method. If the solution is insufficient that does not match the practices or the AI method does not provide an adequate solution, the second level of the algorithm is deployed which suggests using combined AI methods. Combining different AI techniques is, in fact, the current trend in engineering problems and can yield high-performance solutions. For this purpose, the algorithm asks the information regarding the type of engineering design application, the designer needs, and another artificial intelligence method that can be combined with the method chosen at the first level. If the designer needs design support or decision-making, a combination with expert systems is recommended. If the designer's need is an inspiration and/or concept creation, the question "Can data be produced?" is asked again and directed to data-oriented approaches. On the other hand, in cases where data-driven methods do not provide sufficient solutions, the designer is recommended to try rule-based systems or genetic algorithms to make new combinations. It should be noted that the process of determining the AI method here is based on previous AI-enforced design studies covering the last 15 years. The proposed algorithm is especially useful for novice designers.

Future Outlook

The future of AI will most likely be oriented around machine learning and deep learning methods. This is an in-progress process and ease of access to data and increment of computer processing and storage capacity will facilitate this trend. Data is continuously produced by the Internet of Things (IOT), online shopping sites, search engines, and surveys. However, processing, organizing, and labelling this data is expensive and cost time. Text, sound, image, and 3D model databases available can readily be used to solve these problems. Databases of patents contain valuable information for engineering studies (Liu et al., 2020a,b,c; Sarica et al., 2020). Data mining and natural language processing methods are used to process this information. Databases of ontology, on the other hand, help to increase creativity through simulation and association (Liu et al., 2020c). Minimum domain knowledge is required for concept generation, idea, evaluation, and optimization processes with deep learning models. The results can be obtained by determining the relationships among text, images, 3D models, etc. How these relationships are determined is a closed-box problem. Explanation of AI processes via XAI and understanding the underlying mechanisms will increase the reliability of AI systems (Stembert and Harbers, 2019). A large amount of data is required in the training of deep learning methods. Datasets of design (text, images, models, etc.) are still limited.

Design is not just about the aesthetic appearance. The geometric form just includes the dimensional and appearance properties of the product. The function that expresses the purpose for which the product is designed and the behaviour that represents the relation of the product with the user and the environment should be considered as a whole for a good design. Image processing-based approaches generally work on product forms. Functional and behavioural characteristics should also be taken into account for applicable designs (Oh et al., 2019; Shu et al., 2020).

Future studies should focus on improving the model performances that are significantly influenced by the determination of ideal hyper-parameters. In addition, the rather long training periods and data preparation processes should be shortened. To shorten long training times, complete and noise-free design datasets created under suitable conditions are required. Another research topic is training deep learning models with a small amount of data.

7. Conclusions

This article provides a comprehensive overview of current and common AI methods and applications in engineering and product design. AI-powered design applications in the last 15 years were examined through a systematic literature review. The methods and techniques suitable for the design process/problem have been identified, classified, and critically evaluated. An AI method selection algorithm is then proposed based on successful applications reported in the open literature, especially for novice designers. Although quite a lot of publications involving AI-powered design have been reviewed and included in this article, by no means this is complete, and AI applications in design are not limited to this review.

AI methods save time and effort by enabling repetitive design, analysis, and redesign processes to be performed independently. Especially, thanks to the developments in Computer vision technology, the tasks of creating and evaluating 2D and 3D design concepts have been enormously simplified. It has also been cost-effective compared to traditional design methods as more precise results can be achieved in a shorter time. Furthermore, it helped to avoid bias and over-commitment to an idea when generating and evaluating design options allowing designers to reach the global optimum in a wide range of solution areas. The AI methods are objective and achieve similar results under the same conditions. Another advantage of the AI-enforced design is that unlike human-centred design which relies on the designers' limited accumulation of knowledge fading away over time, the knowledge and skills learned in AI never vanish and can be recalled at any time. Although the tasks of the human designers are gradually transferred to artificial intelligence with the developing technology, the AI, in the authors' opinion, will never be self-sufficient for any given creative design task, and the human designers will always be involved in the decision-making process with their knowledge, experience, sensory and intuitive abilities as there is always an error margin in AI produced results.

The choice of AI method in design problems depends on the nature of the problem and the amount of data. Classical AI techniques (artificial neural networks, fuzzy logic, expert systems, and genetic algorithms) have long been used in the evaluation, optimization, and decision-making of design concepts that require expert knowledge and experience. On the other hand, the popularity of data-based methods has been increasing in recent years due to the increase in computer processor speed and storage capacity, the creation of labelled/unlabelled data sets, and the ease of access to huge data. The development of modern data-driven methods has led to a dramatic increase in design research such as inspiration, concept creation, and 3D model creation. The increase in design inspiration and concept creation efforts in recent years is directly proportional to the availability of large amounts of data. These methods started a new era in the design process with their ability to process huge data that human capacity is insufficient. In other words, data-based methods replace knowledge and experience-centred approaches.

There are also some disadvantages to using AI methods in design. The first of these is an AI algorithm developed for a particular design problem that is generally not applicable to other design problems because it lacks heuristic capabilities. Therefore, the ability to adapt to changing conditions is poor, except for some deep learning models. On the other hand, training deep learning models requires a lot of data and high-capacity computers. In addition to all these, many AI applications contain 'black box' solution processes that cannot be fully explained by users/designers. Understanding the causality behind AI decisions provides more reliable and interpretable design solutions. For this purpose, it is foreseen that interpretable and high-performance Explainable Artificial Intelligence (XAI) methods will become widespread in design applications. In the design space, future work should be on AI systems that aim to develop designs with improved form, function, and behaviour. In particular, rapid prototyping techniques that significantly reduce production constraints and simplify design iterations can be incorporated into AI processes. Thus, it will be more feasible for the initial concepts to become functional products in a much shorter space of time. Virtual Reality (VR) and Augmented Reality (AR) technologies can also be integrated into the design process to increase the efficiency and performance of generative design models.

CRediT authorship contribution statement

Nurullah Yüksel: Conceptualization, Methodology, Writing – original draft, Visualization, Formal analysis, Investigation. **Hüseyin Rıza Börklü:** Conceptualization, Methodology, Writing – review & editing, Visualization, Supervision. **Hüseyin Kürşad Sezer:** Conceptualization, Writing – review & editing, Visualization. **Olcay Ersel Canyurt:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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